

EXPLORATION OF ANOMALOUS GLACIERS BEHAVIOR IN RESPONSE TO CLIMATIC VARIABILITY IN THE KARAKORAM REGION



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Abstract

My focus in this thesis was to provide a more thorough understanding of the ‘Karakoram Anomaly’ unless noted cause, the Karakoram glaciers exhibit steady mass or expand in the western and central Karakoram contrasting with the retreat in the eastern Karakoram and respond differently to the climatic conditions compared to glaciers in the world, a phenomenon defined the termed Karakoram Anomaly. The absence of long-term ground-based monitoring of climatic variables and glaciers observations in addition to complex terrain, remote location, and harsh climatic patterns pose a serious challenge to find a precise explanation for anomalous behavior of glaciers and their response to the prevailing climate variability.

The study’s general approach to explore the glaciers behavior was to compare a high-resolution geodetic glaciers data set with climate variables to identify the mass balance condition of glaciers at different elevations and relationship with climate signatures (referred here as Karakoram Climate Response Anomaly - KCRA). Irrespective of changing elevations, this study observed and analyzed that the mass balance of all glaciers is stable and the glaciers at the higher elevations are surging from the terminus in the western and central Karakoram under the unique climate warming over the twenty-first century with slight reduction in snowfall. This study found that the mountainous glaciers land (topography) and their north facing aspect are the main causes of their stability despite prevailing warming climate signatures. Expanding on this work, this study analyzed the future implications under IPCC AR6 (Intergovernmental Panel on Climate Change Sixth Assessment Report) scenarios of climatic variables by using the projected CMIP6 (Coupled Model Intercomparison Project Phase 6) ensemble models three SSPs (Shared Socio-economic Pathways) including (low emission) SSP1-2.6, (medium emission) SSP2-4.5 and (high emission) SSP5-8.5 that depict Karakoram glaciers sustenance difficult in the near future, specifically the western Karakoram experiences rising projected temperature across all scenarios coincide with the drier conditions and reduced precipitation in the recent coming decades. Study findings suggest that regular field observations of glaciers mass balance and climate conditions should be required to further underline the glacier response to climate variability.

Chapter 1

Introduction

The Karakoram region stands as a remarkable geological marvel, situated amidst the Himalayas, Hindu Kush, and Pamir formidable ranges. Spanning across the parts of Pakistan, India, and China, this rugged region is renowned for its highest peaks, deep valleys, and, notably, its vast glacier cover. This introductory chapter sets the stage for a comprehensive exploration into the glaciers' anomalous behavior within the Karakoram region in response to climatic variability.

1.1 Overview of Karakoram region

The Karakoram, sometimes referred to as the "Third Pole," (e.g., Zanchi and Gaetani, 2011; UNDP, 2022; UNEP, 2022) encompasses an expansive area of approximately 450,000 square kilometers, extending across the borders of Pakistan, India, and China. Situated between latitudes 35° N to 37° N and longitudes 74° E to 79° E, some of the highest peaks in the world, including K2 (8,611 meters), Gasherbrum I (8,080 meters), and Broad Peak (8,051 meters), are found in this rugged mountain range (UNESCO, 2010; Chaturvedi and Sakhuja, 2015).

The Karakoram commonly referred to as "home to giant glaciers," has garnered attention from all over the world because of its unique climatic and glacial features (Yao et al., 2012; Pant et al., 2018; Lamchin et al., 2022). The unique topography and climatic conditions of the Karakoram contribute to its distinct glacier characteristics, setting it apart from other mountain ranges (Morton, 2011). The interplay of high peaks, deep valleys, and complex glaciers system makes the Karakoram a focal point for glaciological research, offering valuable insights into glacier dynamics and their response to climate variability.

1.2 Glaciers in the Karakoram

The Karakoram region hosts an extensive network of glaciers, covering a diverse array of glacier types and features. These glaciers range from massive icefields that sprawl across high-altitude

plateaus to small valley glaciers nestled within untamed mountain valleys. These glaciers exhibit diverse features, including surging, debris-covered, and high-altitude glaciers (AZAM et al., 2018; Soheb et al., 2022; Xie et al., 2023). The sheer abundance and variety of glaciers in the Karakoram underscore the need for a nuanced understanding of their behavior and response to changing environmental conditions (Bolch et al., 2008).

The glaciers of the Karakoram cover an estimated area of over 15,000 square kilometers, making them a prominent feature of the landscape. This vast expanse of ice provides millions of people inhabiting the surrounding regions with essential water resources (Gardner et al., 2013). The Indus, Shyok, Hunza, and Shigar river basins receive a substantial amount of their water flow from glaciers, which are fed by the Karakoram's permanent snow and ice (Mukhopadhyay and Khan, 2015). The importance of Karakoram glaciers extends beyond their hydrological significance, as they play a crucial role in sustaining various socio-economic activities downstream. These glaciers support agricultural practices through the provision of meltwater for irrigation, facilitate hydropower generation, and contribute to the overall ecological balance of the region. Consequently, they are integral to the socio-economic fabric of the Karakoram and its surrounding areas.

1.3 Understanding the Karakoram anomaly

The peculiar glacier behavior in the Karakoram region—often referred to as the "Karakoram anomaly", has drawn significant attention from scientists in recent years. This phenomenon challenges conventional notions of glacier retreat observed in rest of the world, where glaciers are generally experiencing significant mass loss and recession (Hewitt, 2005).

The Karakoram anomaly is characterized by the unique response of glaciers in the region, which exhibit either steady mass or even expansion in the western and central Karakoram, contrasting with the widespread retreat observed in the eastern Karakoram and Himalayan glacierized regions (Lhakpa et al., 2022). This peculiar behavior has sparked considerable interest and debate among glaciologists and climatologists seeking to unravel its underlying causes. Several hypotheses have been suggested to elucidate the Karakoram anomaly (Gardelle et al., 2013), each offering insights into different aspects of glacier dynamics and climate interactions. One prominent hypothesis suggests that regional climate patterns, influenced by the complex topography of the Karakoram, play a significant role in shaping glacier behavior. The presence of high-altitude mountain ranges and the interaction of monsoonal and westerly circulation systems create unique climatic conditions that may modulate glacier response to changing environmental variables (Quincey et al., 2009; Bolch et al., 2012).

Furthermore, the influence of debris cover on glacier dynamics has been identified as another potential factor contributing to the Karakoram anomaly. Debris-covered glaciers, common in the region, exhibit distinct thermal characteristics that can insulate the ice beneath, influencing melt rates and mass balance dynamics (Minora et al., 2016; Dobrev et al., 2017). The presence of debris cover may act as a protective layer, shielding glaciers from rapid mass loss and contributing to their resilience in the face of changing climatic conditions (Mölg et al., 2018; Mayr and Hagg, 2019; Huo et al., 2021). Additionally, the complex dynamics of individual glaciers within the Karakoram further contribute to the anomaly. Variations in glacier geometry, flow dynamics, and internal processes can result in heterogeneous responses to climate forcing, leading to spatially diverse patterns of glacier behavior across the region.

Recent studies have highlighted the contrasting responses of Karakoram glaciers to climatic conditions compared to glaciers in other regions of the world. While many glaciers worldwide are experiencing widespread retreat and mass loss, Karakoram glaciers exhibit a more varied response, with some glaciers maintaining stability or even advancing in certain areas. This divergence underscores the demand for a nuanced understanding of regional climate processes and their interactions with glacier dynamics.

1.4 Glaciers' mass balance and variability

At the core of understanding glacier dynamics lies the concept of glacier mass balance. Figure 1.1 shows the mass balance of glacier represents the net increase or decrease of ice mass during a

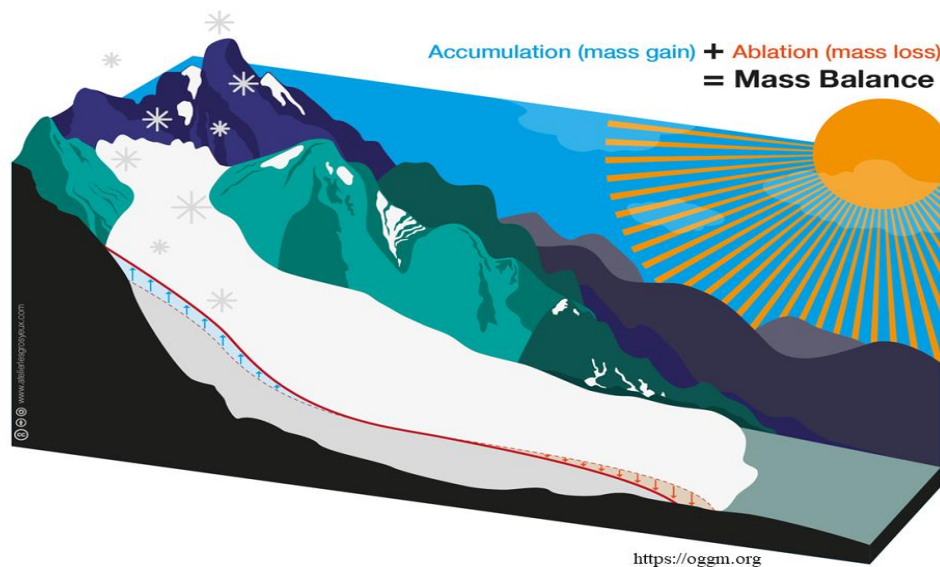


Figure 1.1: Concept of mass balance of glacier.

certain time period, determined by the intricate interplay of snow accumulation, melt, and ablation processes (Cuffey and Paterson, 2010).

Glaciers consist of two primary zones (Menzies and Hughes, 2002; Muhammad and Tian, 2016).

Accumulation Zone

In the accumulation zone, snowfall accumulates over time, gradually compacting into ice and contributing to the glacier's mass.

Ablation Zone

In the ablation zone, ice loss occurs through melting, sublimation, and calving, resulting in a net reduction in glacier mass.

Variations in mass balance offers valuable information on the glacier health and response to climate variability (Hock, 2005; Huss, 2012). In general, mass balance varies from glacier to glacier.

Positive mass balance. It happens when accumulation exceeds ablation, leading to a net gain of ice mass. This typically occurs in regions where snowfall is abundant and temperatures remain consistently low, leading to glacier growth.

Negative mass balance. Conversely, it occurs when ablation surpasses accumulation, leading to a net loss of ice mass. Negative mass balance is often associated with rising temperatures, reduced snowfall, and increased melting rates, contributing to glacier retreat.

In some cases, glaciers may exhibit a **balanced** or **zero mass balance**, where accumulation equals ablation. This delicate equilibrium reflects a state of dynamic equilibrium, where the glacier's size remains relatively stable over time. However, even balanced glaciers are susceptible to climate change, as shifts in temperature and precipitation patterns can disrupt this delicate balance (Pfeffer et al., 2014), leading to fluctuations in mass balance and glacier behavior.

To understand the variations in mass balance is crucial to estimate the future glacier trends and evaluate the influence of climate change on glacier systems. By tracking the mass balance changes over time, to obtain insights into how glaciers respond to shifting climatic conditions. These insights can then be used to improve conservation efforts and adaptation strategies in places that are glacierized.

1.5 Equilibrium Line Altitude (ELA) assessment

Equilibrium Line Altitude (ELA) serves as a fundamental metric in glacier mass balance studies, providing critical insights into glacier response to climate variability. ELA defines the elevation

at which snow accumulation equals ablation over a specific period, delineating the boundary between the glacier's accumulation and ablation zones (Hock, 2010; Braithwaite, 2015). In the context of the Karakoram, ELA analysis offers a valuable tool for assessing glacier health and unraveling the complexities of the Karakoram anomaly. By analyzing changes in ELA over time, researchers have identified spatial and temporal patterns of glacier response to climatic variability (Rupper et al., 2009). ELA estimation involves a combination of numerical modeling, remote sensing methods, and field observations. Field-based methods often include the installation of snow pits and ablation stakes to determine snow accumulation and melt rates at different elevations along a glacier (Kaser et al., 2003; Karpilo, 2009; Stumm et al., 2017). Remote sensing methods, including satellite imagery and aerial photography, assist to map glacier extent and identify changes in surface features associated with ELA fluctuations (Rabatel et al., 2013; A. E. Racoviteanu et al., 2019).

1.6 Measuring glaciers mass balance through ELA

The determination of glacier mass balance through ELA techniques involves combining field measurements include conducting meticulous surveys and ablation stake measurements and snow pit analysis techniques, and remote sensing data such as satellite imagery and digital elevation models (DEMs) to track changes in glacier extent over time (e.g., Bamber and Rivera, 2007; A. Racoviteanu et al., 2008; Rabatel et al., 2008; Paraj, 2015).

Figure 1.2 shows the mass balance of glaciers estimate by analyzing changes in ELA position over time. If ELA rises, it indicates increased melting and a negative mass balance, signifying more ice loss than gain.

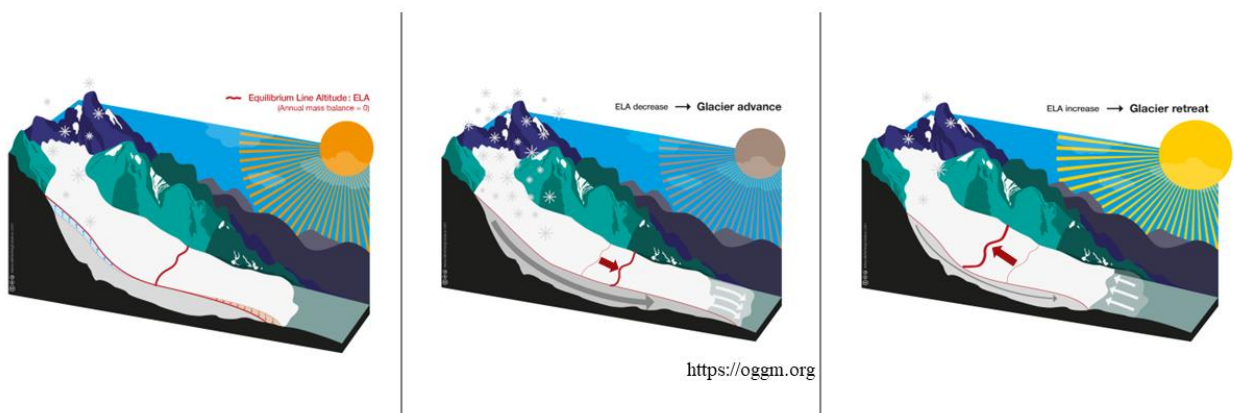


Figure 1.2: Identifying glacier mass balance through ELA analysis.

Conversely, a lowering ELA suggests reduced melting and a positive mass balance, indicating more ice accumulation than loss (Rabatel et al., 2012; Mernild et al., 2013; De Angelis, 2014; Six and Vincent, 2014). ELA analysis helps assess glacier health and comprehend the consequence of climate change on glacier dynamics. Monitoring ELA variations provides valuable insights into the drivers of glacier mass balance and informs predictions of future glacier behavior.

1.7 General glacier responses to climate change

Glaciers around the world exhibit diverse responses to climate change (Huss and Hock, 2018; IPCC, 2019), driven by a complex interplay of environmental factors. While individual glaciers may respond uniquely to local conditions, certain general trends can be observed across glacierized regions (Sagredo and Lowell, 2012; Rabatel et al., 2013). Changes in atmospheric circulation, rising temperatures, and altered precipitation patterns influence glacier dynamics, leading to alterations in mass balance and extent.

The rising of global temperature has a profound impact on glacier dynamics, leading to accelerated melting and retreat. Warmer temperatures enhance the ice melt rate (Scherler et al., 2011; Bliss et al., 2014), particularly at lower elevations and in regions with seasonal snow cover. This intensified melting contributes to negative mass balance and leads to overall glacier shrinkage over time. The precipitation patterns shift, including changes in the timing, intensity, and form of precipitation, can influence glacier mass balance (Vizcaíno et al., 2010; Marzeion et al., 2014; Østby et al., 2017; MORTIMER et al., 2018). Changes in snowfall patterns affect the glaciers' accumulation zone, variability in precipitation can also lead to variations in the magnitude and frequency of glacier runoff, affecting downstream water resources. The alteration in atmospheric circulation patterns, such as shifts in prevailing wind directions and the occurrence of harsh weather events (e.g., Ganopolski and Rahmstorf, 2001; Vuille et al., 2008; Zhu et al., 2015; T. Mölg et al., 2020), can influence glacier dynamics. Changes in atmospheric circulation can affect regional temperature and precipitation patterns, affecting the distribution of snowfall and melt across glacier surfaces. Additionally, variations in atmospheric circulation can impact the transport of moisture and aerosols, which can alter the mass balance of glaciers through processes such as dust and black carbon deposition. The identification of these general responses of glaciers to climate change is critical for forecasting future glacier behavior and evaluating the related risks.

Objectives of the study

The studies on high mountain Asian climatic changes yield contradictory results, some lack significance due to incomplete spatial and temporal coverage. The meteorological stations show unchanged seasonal and annual temperature and precipitation patterns (Shrestha et al., 2000; Bhutiyani et al., 2007), while observed considerable warming at lower elevations based on gridded and satellite data (Gautam et al., 2009; Immerzeel et al., 2009). The causes of glacier expansion include deformable bed theory, thermal regime, hydrological control, and critical mass (Cui et al., 2021). Although it is still unknown how much, these snow and ice formations are probably impacted by the region's shifting climate (Immerzeel et al., 2010). While several explanations have emerged, a comprehensive understanding of the anomaly remains elusive. However, a clear consensus on the benefits and drawbacks of glacier expansion in the Karakoram region remains unclear.

This study presented the behavior of different altitudinal Karakoram glaciers in relation to their altitudinal climatic conditions. The approach to this problem involved addressing gaps from previous literature and using satellite data to explore the variations in glaciers mass balance at different elevations and to identify the historical climatic change signals across space and time. This study compared the mass balance of altitudinal glaciers with the climatic variables and discussed the glaciers relationship with climate and try to find the relevance unnoted cause behind the glaciers behavior. This study also presented the future implications of Karakoram glaciers according to projected climate using the CMIP6 models under IPCC AR6 scenarios.

Chapter 2

Literature Review

Initially, Hewitt (2005) observed the ‘Karakoram anomaly’ as a sub-regional positive glacial mass balance trend. This anomaly, characterized by balanced or slightly positive mass gains, has intrigued scientists since its identification in the Karakoram region after 1970 (Cogley, 2011; Wester et al., 2019). It stands in direct contrast to the global negative glacier mass balance trend observed since 1850 (Copland et al., 2011; Cogley and Waltham, 2012; Brun et al., 2017; IPCC AR6, 2022). Despite early observations by 19th-century explorers (e.g., Desio, 1954), the persistence of this phenomenon has fueled significant scientific discourse. Numerous glacier mass balance studies have quantified the Karakoram anomaly, shedding light on its complex dynamics and contributing to our understanding of regional climate processes (e.g., Bolch et al., 2012; Gardelle et al., 2012; Jacob et al., 2012; Kääb et al., 2012; Pratap et al., 2016; Bolch et al., 2017; Zhou et al., 2017). These studies highlight the importance of considering regional-scale factors, including debris cover, topography and atmospheric circulation patterns, in explaining the unique behavior of Karakoram glaciers within the broader term of global climate change.

Fowler and Archer (2006) identified contradictory indications of changing climate in the Upper Indus Basin, particularly in relation to summertime temperatures over the Karakoram region. The study revealed a paradoxical trend of cooler summertime temperatures observed in the Karakoram, contrary to the warming trends observed in other regions globally. This cooling trend, identified through analysis of temperature data, suggests a unique climatic response in the Karakoram.

Janes and Bush (2012) investigated the role of climate change and atmospheric dynamics in influencing precipitation trends and the potential fate of glaciers in the Karakoram region. They discussed the complex interactions between westerly winds originating from the Mediterranean

and Caspian Seas and monsoon winds from South Asia, originating from the Indian Ocean. These atmospheric dynamics contribute to the intricate precipitation patterns observed in the Karakoram, where moisture laden westerly winds interact with the moisture rich monsoon winds. The convergence of these wind systems results in temporal and spatial variations in precipitation across the region, influencing glacier mass balance and hydrological regimes.

Kapnick et al. (2014) study focusing on the unique seasonal cycle of snowfall in the Karakoram region and its implications for the snowfall response to global warming in the twenty-first century. The research revealed that the Karakoram seasonal cycle is characterized by a distinctive pattern that provides protection against the decline trend in annual snowfall associated with global warming. Unlike the Himalayas, where snowfall is highly sensitive to warming trends, the Karakoram region exhibits a less pronounced response to rising temperatures due to its unique seasonal cycle. The study found that the Karakoram experiences a delayed peak in snowfall during the spring months, which coincides with the onset of warming temperature. This delayed peak in snowfall acts as a buffer against the impacts of warming, mitigating the decline in annual snowfall observed in other mountainous regions. The regional climatic factors are important, such as the seasonal timing of snowfall, in understanding the snowfall behavior to global warming in mountainous regions.

The study conducted by Mölg et al. (2014) focused on the mid-latitude westerlies impact on glacier variability in monsoonal High Asia, including the Karakoram region. It revealed that the mid-latitude westerlies are important in driving abnormal changes observed in glacier dynamics in this region. Through a combination of observational data and modeling techniques, the study demonstrated that variations in the strength and direction of the mid-latitude westerlies influence atmospheric circulation patterns, precipitation regimes, and temperature distributions in monsoonal High Asia. These atmospheric changes, in turn, impact glacier dynamics and mass balance, leading to alterations in glacier extent and behavior. The research highlighted the complex interactions between regional atmospheric circulation systems and glacier variability, emphasizing the importance of considering large-scale climatic drivers in understanding glacier dynamics in the Karakoram and surrounding regions.

An extensive study was conducted by Paul (2015) on the western Karakoram (Hunza basin) particularly. He utilized geodetic methods to investigate glacier surge and flow dynamics in the western Karakoram region, revealing a balanced or slightly positive budget since approximately 2000. Through the analysis of animated satellite image sequences, the study provided insights into the intricate dynamics of glacier flow and surge events. The findings highlighted the

resilience of glaciers in the western Karakoram, which have maintained a stable or slightly positive mass balance despite global trends of glacier retreat.

Bhambri et al. (2017) study focusing on surge-modified and surge-type glaciers in the Karakoram region, shedding light on the function of glacier meltwater in influencing regional climate processes. This study highlighted the significance of glacier surges in modulating glacier dynamics and hydrological regimes in the Karakoram. They revealed the importance of surge-type glaciers as significant contributors to glacier mass balance and runoff generation in the region.

Forsythe et al. (2017) conducted a comprehensive study investigating the intricate relationship between the Karakoram vortex, regional atmospheric circulation variability, temperature anomalies, and glacial melt in the Karakoram region. Their research revealed the complex interplay between the Karakoram vortex and westerly jet position, which exerts significant influences on temperature patterns and streamflow anomalies in the region. The study found that South Asia's winter temperature exhibits a uniform response to regional circulation metrics, while the summer temperature of the Karakoram region displays distinct variability compared to other Himalayan regions. This variability is attributed to the seasonal contraction of the Karakoram vortex during interactions with the South Asian monsoon. The research highlighted the critical role of regional atmospheric circulation variability in driving temperature anomalies and glacial melt in the Karakoram, emphasizing the need for further investigation into the mechanisms governing these interactions.

Farinotti et al. (2020) predicted the unique glaciers behavior in the Karakoram and its implications for downstream hydrology and hazards. The shifts in streamflow patterns have led to alterations in the frequency and magnitude of glacial lake outburst floods (GLOFs), posing increased risks of flooding downstream. The study highlighted the significant impact of these changes on populations and infrastructure, including hydropower and large-scale irrigation systems, in the region. The findings underscored the importance of proactive measures to mitigate the risks associated with changing streamflow patterns and glacier-related hazards in the Karakoram region, particularly in the climate change context and water scarcity.

There are two comprehensive studies conducted by the same authors on the assessments of glacier dynamics in the Karakoram region, focusing on the Hunza basin in western Karakoram and the Shigar basin in central Karakoram. The researchers aimed to comprehend the behavior of glaciers to climatic changes and factors influencing glacier response in these regions.

In the western Karakoram, Ali et al. (2021a) utilized satellite imagery and remote sensing techniques to analyze glacier changes from 1990 to 2018. The results revealed the slightly positive or balanced mass budget in the region, despite a decreasing trend in temperature and increasing precipitation. This balanced mass budget suggests a relatively stable glacier condition in the western Karakoram, challenging global trends of glacier retreat observed elsewhere. Meanwhile in other study, Ali et al. (2023) investigated glacier changes in central Karakoram from 1973 to 2020 used a combination of field surveys, remote sensing data, and climate records. The findings indicated glacier shrinkage due to reduced precipitation and rising temperatures, yet recent glacier advances were observed, likely influenced by increased winter and autumn precipitation. These contrasting trends highlight the complex interplay of climatic and topographic factors in the Karakoram region.

The studies contribute valuable insights into the response of Karakoram glaciers to climatic change and the factors driving glacier behavior in both the central and western Karakoram. By combining remote sensing techniques, field surveys, and climate data analysis, the researchers shed light on the unique dynamics of glaciers in these regions, contributing to our knowledge of glacier behavior to climate variability.

The findings from Nie et al. (2021) study revealed that the dynamics of individual glacier surges are out of phase, suggesting a limited impact of climatic conditions on surge behavior. This highlights the complex nature of glacier dynamics in the Karakoram and underscores the importance of considering various factors, such as glacier morphology and internal processes. The study contributes valuable insights into the mechanisms driving glacial change in the region and the potential consequences for downstream water resources and ecosystems.

In another study, Dong and Ming (2022) attempted to identify the variability and seasonality of the snowfall to total precipitation ratio across High Mountain Asia, including the Karakoram region, using simulations of the GFDL (Geophysical Fluid Dynamics Laboratory) High-Resolution AM4 model. The outcomes revealed a downward trend in the snowfall to total precipitation ratio in the recent past, primarily driven by a decrease in snowfall. Despite an increase in total precipitation, the decline in snowfall has contributed to the unusual behavior observed in the region. The study also underscores the importance of considering changes in precipitation phase and distribution in understanding the response of mountainous regions to climate change.

Another cause behind the Karakoram anomaly described by the study of Javed et al. (2022). The western disturbances (WDs) might be in governing the Karakoram anomaly, particularly in

relation to winter precipitation and seasonal snowfall. The study revealed that WDs, characterized by extratropical cyclonic storms originating in the Mediterranean region, play a substantial role in bringing winter precipitation to the Karakoram region. The winter precipitation contributes a substantial amount of seasonal snowfall, which has been identified as one of the main factors behind the development of the Karakoram anomaly. The research examined the recent revival of WDs and its potential implications for the Karakoram anomaly, highlighting the importance of considering synoptic-scale atmospheric processes in understanding glacier dynamics and variability in the region.

The dynamics of WDs involve the interaction of polar and subtropical air masses, leading to the development of cyclonic circulations and frontal systems. As WDs move eastward, they encounter the complex topography of the Karakoram and interact with local atmospheric conditions, influencing the distribution and intensity of precipitation. The revival of WDs in recent years has drawn attention to their role in governing the Karakoram anomaly, highlighting the necessity of further research to comprehend the mechanisms driving their behavior and their implications for regional climate variability.

Chapter 3

Methodology

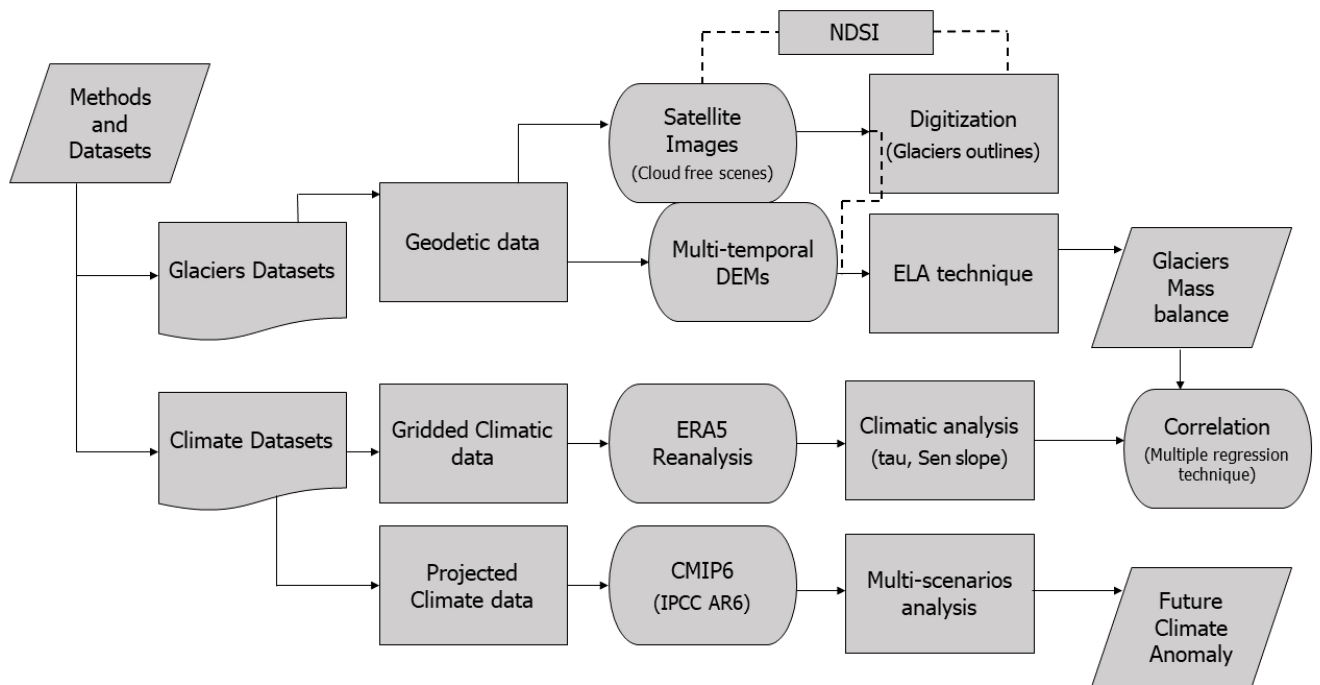


Figure 3.1: Flowchart of methodology.

3.1 Selection of study area

This study separated the two basins from the total three basins of the Karakoram region to explore the geographical variations in the glaciers behavior (Figure 3.2 and summarized in Table 3.1). Basin 1, the Hunza Basin, referred to as collective western Karakoram region and Basin 2, the Shigar Basin, referred as central Karakoram region, were chosen for their significance in glacier dynamics and climatic conditions. The Shyok basin covers the eastern Karakoram region was not

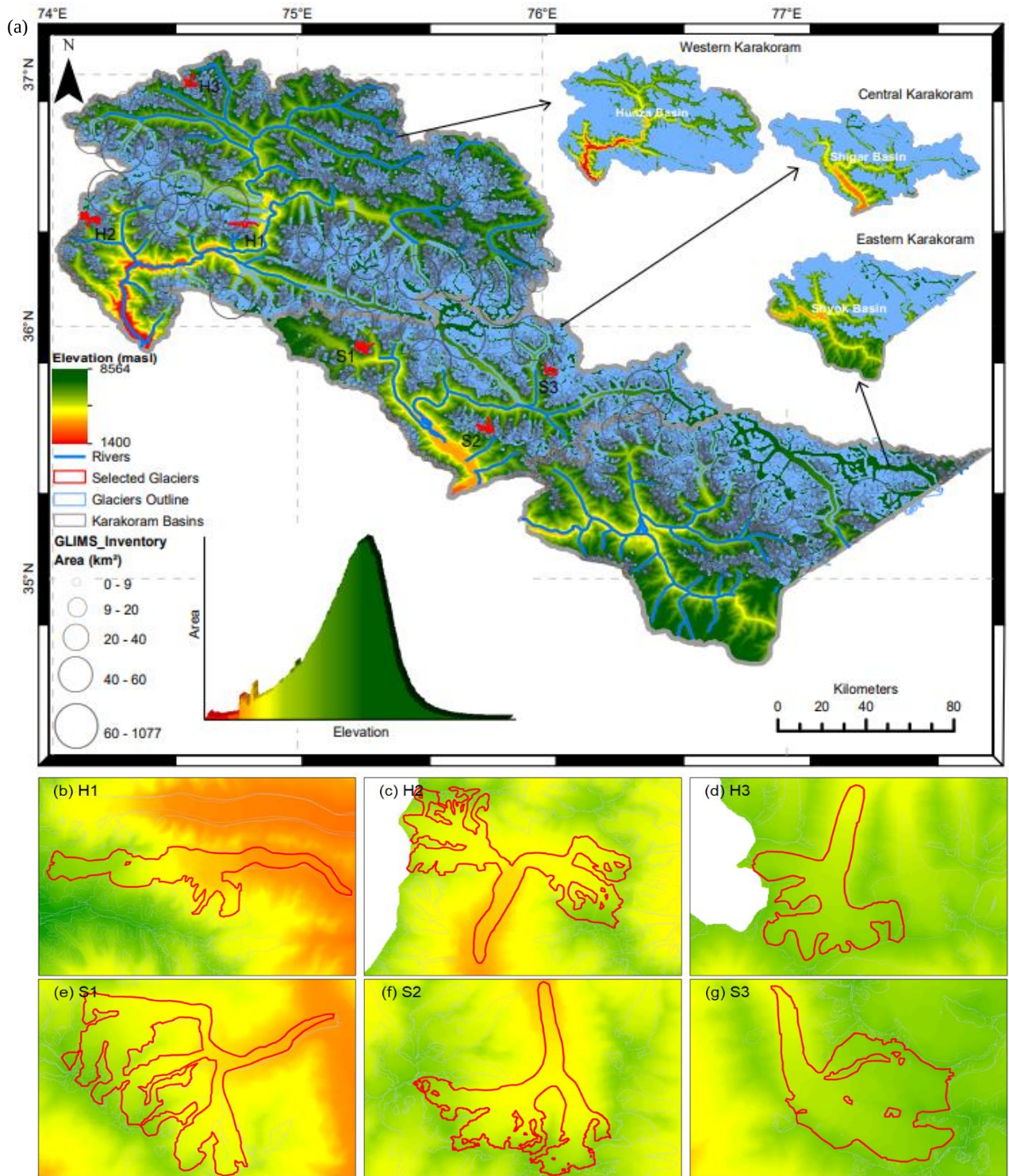


Figure 3.2: Mapped region of interest with hypsometry and unidentified selected glaciers. a, SRTM DEM based topography shown the different elevation range up to 8000 m a.s.l. of the Hunza, Shigar and Shyok basins of Karakoram region as specified, GLIMS Inventory given glaciers outline. Hypsometry graph obtained by area distribution across each specified elevation. b-d, three glaciers' GLIMS defined outline of Hunza basin, lie at approx. 3000m, 3500m and 4600m elevations respectively and represented by given names (according to Table 3.2) in this study as H1, H2, and H3 in Figure1a. e-g, specified as S1, S2 and S3 in figure1a are three Shigar basin glaciers lie at corresponding elevations of Hunza glaciers.

included in the study, due to no particular anomaly found there in scientific literatures (Ougahi et al., 2022). All basins had mean elevations above 4500 m a.s.l. (Table 3.1) and experienced different levels of climatic and mass balance condition of glaciers.

Both study basins converged in topographic variability than the other High Mountain Asia regions. The central Karakoram glaciers were either stable or advancing (Dehecq et al., 2019), about 50 % from snout (Scherler et al., 2011). On the other hand, the western Karakoram glaciers were stable and slightly advance (Attaullah et al., 2022) condition similar to central Karakoram. Meanwhile, some glaciers in this region displayed heterogeneous mass balance condition (Shrestha et al., 2015; Wu et al., 2020). Dividing both basins into different elevations in Figure 3.2 to broaden the knowledge regarding glaciers' behavior at various altitudes.

Table 3.1: Summary of observed glaciers behavior and climatic trends in Karakoram basins. Summarizing information provided for available observation studies of glaciers mass balance and meteorological variables i.e., temperature, precipitation and snowfall given for different time periods and using a variety of methods with latitude and longitude range. Trends are given as positive, negative and null (according to the scientific literature).

Karakoram Basin	Latitude Range	Longitude Range	Period	Method	Elevation (m)	Mass balance	Climatic Trend				References
							Glaciers behavior	Temperature	Precipitation	Snowfall	
Hunza	[35,37] N	[74,75] E	1990-2018	Satellite & In-situ	1394-7885	Ø	-	+			Ali et al. 2021
Shigar	[35,36] N	[74,76] E	1973-2020	Satellite & Gridded	2775-8611	+,Ø	+	-	+		Ali et al. 2023
Shyok	[33,36] N	[76,79] E	1990-2014	Satellite & Gridded	2387-7724	-,+,Ø	+	-		Ø	Negi et al. 2021

3.2 Selection of altitudinal glaciers

This study used the inventory of GLIMS RGI version 7 from the National Snow and Ice Data Center (RGI Consortium, 2023) to classify the elevational glaciers in both basins of Karakoram. Only three unknown small mountainous glaciers (names not identified yet, only with GLIMS IDs were available in Table 3.2) were chosen from each basin with correspond elevation difference, which were given names as H1, H2, and H3 lie at 2,991 m, 3,597 m, and 4,628 m a.s.l. in the

Hunza basin respectively, similar in Shigar basin; S1, S2, and S3 lie at 3,380 m, 3,696 m, and 4,609 m a.s.l. with approximately area within 10-14 km² range. These glaciers predominantly exhibited clean ice-covered surfaces, with some showing a small portion of debris-covered ice on terminus at specific periods. RGI version 7, an updated worldwide glaciers inventory of GLIMS released in September 2023, was used to identify and acquire the base outlines of glaciers and used in the analysis of spatial mass balance difference.

Table 3.2: List of selected glaciers, chosen from RGI version 07 used in this study with GLIMS ID, given area and the minimum, mean and maximum elevation. Three glaciers from each selected basin lie on different elevations in the Karakoram region.

No. of Glaciers	Glaciers	Basin	RGI-v7 GLIMS ID	Area (km ²)	Min. Elevation (m)	Mean Elevation (m)	Max. Elevation (m)
1	H1	Hunza	G074798E36411N	9.2	2991	4275	6004
2	H2	Hunza	G075284E35921N	12.7	3597	4631	5687
3	H3	Hunza	G074152E36425N	10.5	4628	5209	5736
4	S1	Shigar	G075783E35611N	13.1	3380	4668	5826
5	S2	Shigar	G074565E36976N	11.9	3696	4707	5670
6	S3	Shigar	G076032E35819N	11.6	4609	5307	5850

3.3 Geodetic data

In this study, geodetic data played a crucial role in mapping glacier outlines and analyzing glacier characteristics over time. Temporal satellite imagery was utilized to obtain cloud-free scenes of Landsat-7ETM+, Landsat 5TM (30m resolution) were acquired from reputable source such as [EarthExplorer \(usgs.gov\)](https://earthexplorer.usgs.gov/), and Sentinel 2 (10m resolution) images from the Copernicus Open Access Hub (<https://dataspace.copernicus.eu/>) to map the outlines of selected glaciers for the

years 2000, 2010, and 2022, respectively. The selection of scenes was carefully executed, with a specific month window (September and October) chosen based on the transition between hydrological years (i.e., ending and starting of hydrological year respectively), ensuring consistency in data acquisition across different time periods.

To facilitate hypsometry calculations and extraction of contours for estimation of ELA and AAR (Accumulation Area Ratio), the study also employed multi-temporal Digital Elevation Models (DEMs). The SRTM V3 Global 1 arc second DEM (NASA Shuttle Radar Topography Mission with 30m resolution) covering the time period of 1991-2000 was accessed through Google Earth Engine (GEE) (Appendix A) for the initial phase of the study. Subsequently, the ALOS-PALSAR DEM (Advanced Land Observing Satellite Phased Array Type L-band SAR with 12.5m resolution) for the time period of 2001-2010 was acquired from the ASF (Alaska Satellite Facility) Data Search portal (<https://search.asf.alaska.edu/>). Lastly, the AST14-DEM (ASTER Digital Elevation Model with 30m resolution) covering the time period of 2011-2022 was obtained from the NASA Earthdata Search portal (<https://search.earthdata.nasa.gov/>).

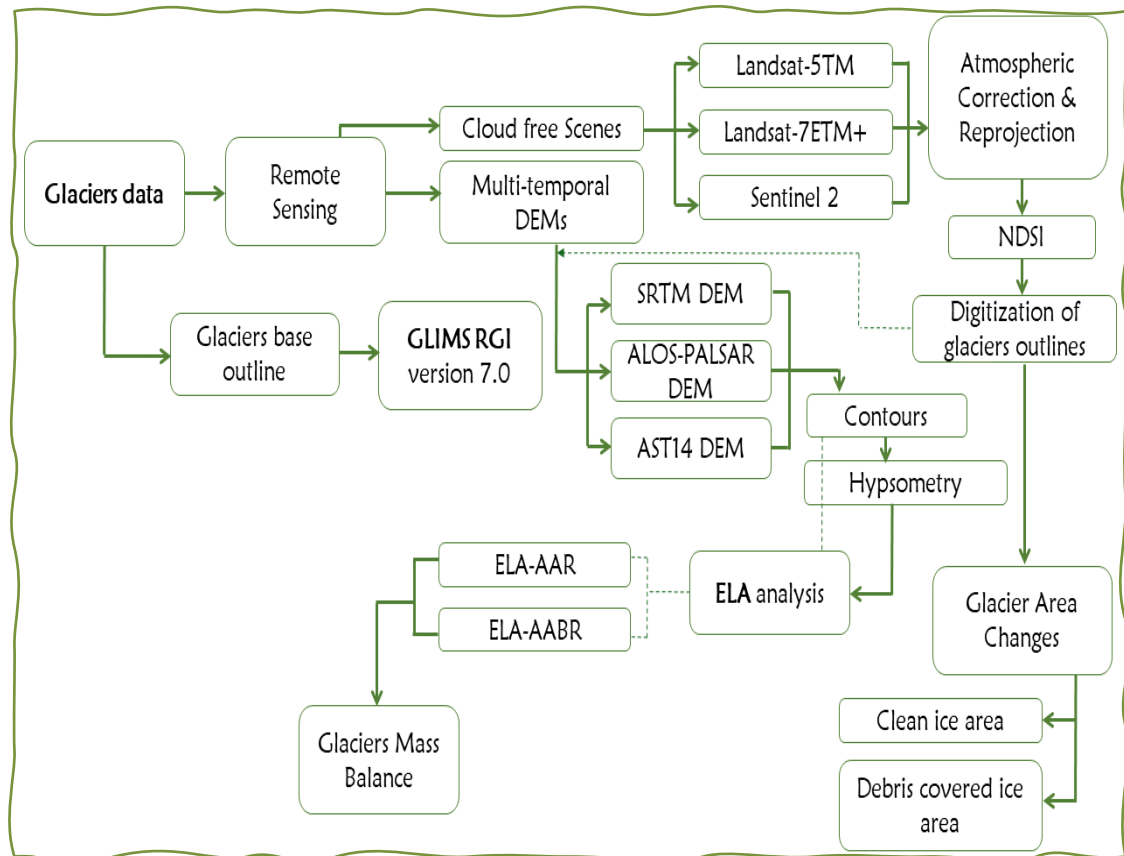


Figure 3.3: Approach of geodetic data to estimate the glaciers' mass balance.

Figure 3.3 comprehensive approach allowed the study to utilize high-quality geospatial data sources and advanced remote sensing practices to accurately assess glacier characteristics and changes over period, contributing to a robust evaluation of glacier dynamics in the study area.

3.3.1 Data processing

This section described the data processing techniques employed to analyze satellite imagery and extract pertinent information regarding glacier outlines and characteristics.

3.3.1.1 NDSI method

The Normalized Difference Snow Index (NDSI) method (Hall and Riggs, 2010) served as a critical tool for distinguishing between snow and other surface types in satellite imagery. Utilizing a standard band combination comprising the green and Shortwave Infrared (SWIR) bands, here applied the NDSI method to the cloud-free scenes obtained in different time series. These bands were specifically chosen to maximize the spectral difference between snow and non-snow surfaces, allowing to accurately delineate glacier boundaries. The NDSI calculation involved computing the normalized contrast between the reflectance values in the SWIR and green bands, effectively highlighting snow-covered areas within the imagery. This process was performed in ArcGIS version 10.8 after reprojecting and applying atmospheric correction to the satellite data, ensuring the accuracy of the results by equalizing the resolution of all time images. By applying the NDSI method, this study was able to effectively identify and digitize glacier outlines, separating debris-covered and clean ice areas for further analysis.

3.3.1.2 Digitization of glaciers outlines

Following the application of the NDSI method, the next step involved digitizing the outlines of selected glaciers temporally. This process facilitated the creation of accurate glacier boundary maps, enabled here to assess changes in glacier extent over time. Leveraging the cloud-free satellite scenes and the delineated glacier outlines, the study utilized ArcMap to digitize the glacier boundaries, ensuring precision and consistency in the delineation process. By digitizing the glacier outlines, the study differentiated between debris-covered and clean ice areas (Ali et al., 2021b), providing valuable insights into glacier morphology and dynamics.

3.3.2 Hypsometric analysis and contours mapping

The study employed DEM data to categorize glaciers into different elevation classes, extracting contours to calculate hypsometry, and determine the cumulative distribution of glacier area at each elevation. Figure 3.2 illustrates the hypsometric analysis curve, showcasing the glaciers

elevation distribution within the Karakoram region. Hypsometric analysis involved studying the elevation distribution within a specific area of interest (AOI) and provides valuable insights into the topographic characteristics of the terrain. Additionally, contour maps played a crucial role in visualizing the terrain morphology and glacier boundaries within the study area. On a topographic map, contour lines indicate lines of equal elevation and give essential information about the slope of the terrain. The generated detailed contour maps accurately depicted the elevation variations across the glaciers.

Moreover, the slope and aspect of glaciers (Negi et al., 2021) were calculated using the ArcGIS statistical toolbox, enhancing our understanding of glacier characteristics and terrain features. This comprehensive approach enabled the study to conduct a thorough analysis of glacier behavior and changes over time, contributing to a better understanding of glacial dynamics in the study area.

3.4 ELA description

The ELA, also commonly referred to as the snowline, marks the altitude on a glacier where the annual accumulation of snowfall equals the amount of ablation (Cogley et al., 2010). It represents the border between the ablation and accumulation zones, where snow melts and accumulates respectively. Determining the ELA was essential for understanding glacier mass balance dynamics and assessing their behavior in response to climate variability. When there are no direct mass balance readings were available for a current glacier, it was frequently necessary to determine the ELA (Osmaston, 2005).

3.4.1 ELA-AAR

ELA-AAR (Accumulation Area Ratio) is a key metric used to assess mass balance of glacier (Bibi et al., 2023) by analyzing the ratio between the area of accumulation and the total glacier area. This ratio provided insights into the relative proportions of the ablation and accumulation zones within a glacier.

$$\text{AAR} = \text{Area of accumulation} / \text{Total glacier area} \quad (3.1)$$

3.4.2 ELA-AABR

The Accumulation Area Balance Ratio (AABR) is another crucial parameter used to assess glacier mass balance. AABR represents the area of accumulation divided by the area within the accumulation zone where mass balance is positive. By incorporating AABR results alongside the

AAR, a more comprehensive understanding of glacier mass balance dynamics was obtained and further support the ELA-AAR results of study, allowing for more accurate estimations of ELA and its implications for glacier health.

3.4.3 Pellitero ELA toolbox

Pellitero et al. (2015) presented a complex toolset in the ArcGIS platform to process the glacier ELA based on AAR and AABR calculations. This toolbox streamlined the ELA determination process by providing efficient algorithms for analyzing glacier characteristics and delineating the boundary between accumulation and ablation zones. Utilizing Python code on reconstructed glacier surfaces derived from DEMs, the toolbox offered a user-friendly interface for researchers to execute ELA calculations and derive valuable insights into glacier mass balance dynamics.

Table 3.3: Summarizing the information of ELA-AAR from the previous literature.

Range and Location of Glaciers	Suggested AAR-Ratio	References
Karakoram glaciers	0.6	Baig et al. 2021
Cirque and Valley glaciers	0.6 ± 0.05	Bakke et al. 2011
(Arctic, Alpine and Asian) glaciers	0.67	Braithwaite et al. 1980
Glaciers larger than 4 km ²	0.64 ± 0.04	Kern et al. 2010
All glaciers (WGI)	0.559 ± 0.09	
Palaeoglaciers	0.67	Pellitero et al. 2015

This study employed the Pellitero given ELA toolbox to determine ELA for >4km² valley glaciers, utilizing the recommended AAR ratio of 0.6 ± 0.05 with a 50m contour interval given in table 3.3. Additionally, utilized correlation coefficient-based linear regression models for mass balance estimation, facilitating a robust analysis of glacier response to climatic variability.

3.5 Gridded observations

Study utilized ERA-5 land monthly gridded products, including temperature, total precipitation, and snowfall datasets for each elevational glacier, spanning the period from 1951 to 2022. The ERA5 reanalysis, developed by the European Centre for Medium-Range Weather Forecasts (ECMWF), symbolizes the fifth generation of atmospheric reanalysis products (Hersbach et al., 2020) and offers high-resolution data with of 0.25 degrees spatial resolution and a temporal resolution of daily gridded data. These gridded products provided comprehensive coverage of long-term climate variables, available from 1950 to present, and were available for download at <https://cds.climate.copernicus.eu/>.

The ERA-5 land monthly gridded datasets offered invaluable insights into the annual and seasonal cycles of climate parameters, allowing for a detailed analysis of climatic conditions affecting elevational glaciers within the study area. The availability of long-term data facilitated a comprehensive understanding of climate dynamics and their potential implications for glaciers mass balance and hydrological processes.

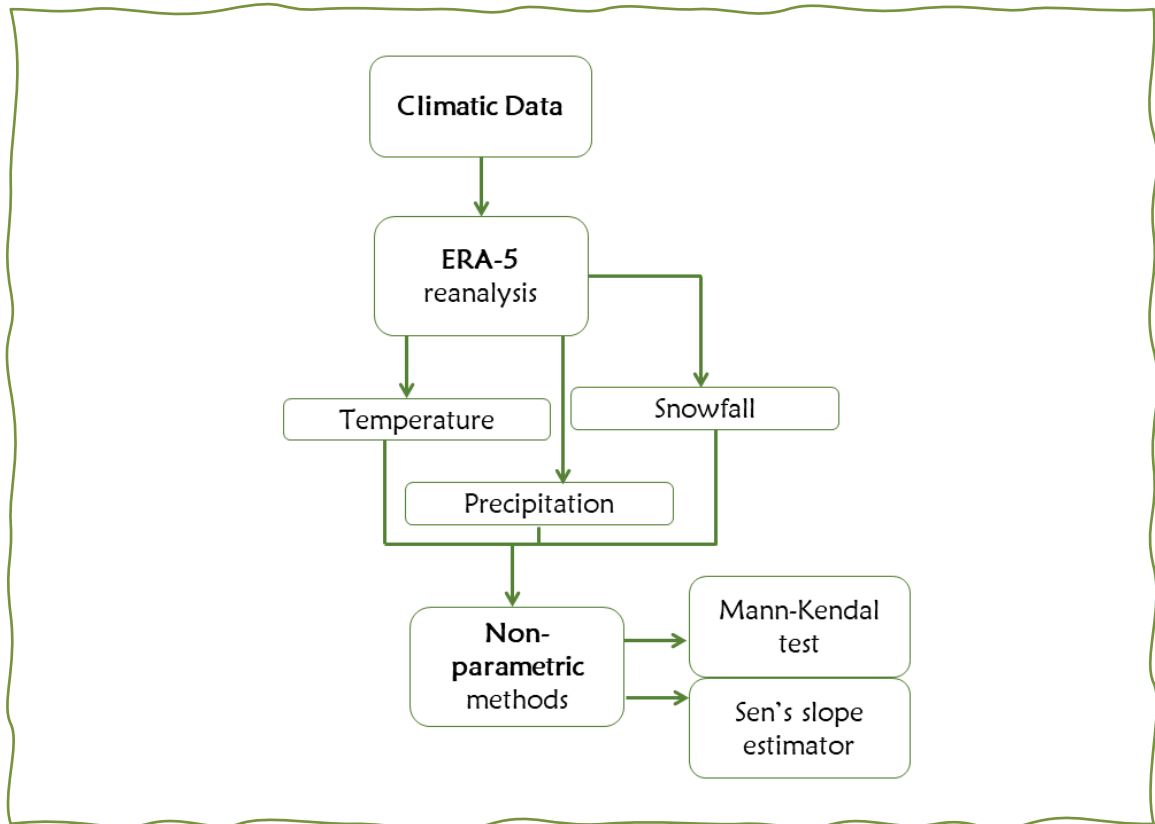


Figure 3.4: Approach of climatic data analysis.

3.6 Climate change signals identification

In the study analysis, employed the Sen's slope estimator and Mann-Kendal test, both recognized non-parametric methods (Figure 3.4), to quantify climate change signals (Hayat et al., 2023). Utilizing the base period spanning from 1951 to 2022, p-values were computed for each climatic variable of interest, enabling the detection of significant trends over time. These methods facilitated the identification of systematic changes in climatic conditions, offering valuable insights into long-term climate trends over the study area.

3.6.1 Mann-Kendal test

The Mann-Kendal test provided a robust statistical framework for detecting trends in climatic variables, allowing to assess the presence of significant upward or downward trends over the study period. By calculating the Kendall tau rank correlation coefficient, the monotonicity of trends in temperature, precipitation, and snowfall was evaluated, providing quantitative evidence of climate change signals.

3.6.2 Sen's slope estimator

Sen's slope estimator complemented the Mann-Kendal test by providing a robust method for estimating the magnitude of trends in climatic variables. By measuring the median of all pairwise slopes among data points, Sen's slope estimator offered a reliable indicator of the rate of change in temperature, precipitation, and snowfall during the study period. This method allowed to calculate the slope of trends, providing additional insights into the magnitude and direction of climate change signals.

The combination of the Mann-Kendal test and Sen's slope estimator provided a rigorous approach for assessing long-term climate trends, contributing to a better understanding of climate variability and its influence on glacier dynamics. Climatic conditions by elevational glaciers were determined by calculating the average climate variable per elevations, followed by calculate the correlation between climate change signals and mass balance by using the Python heatmap() function, furthermore, to evaluated the relationship through linear regression model results.

3.7 C mip6 model projection

To explore future climate scenarios and assess their influence on glacier dynamics, climate projection data of the CMIP6 (Coupled Model Intercomparison Project Phase 6) ensemble models

were utilized. These models form the basis and support the IPCCs AR6 (Intergovernmental Panel on Climate Change Sixth Assessment Report), obtained from the World Bank CCKP (Climate Change Knowledge Portal) 2024 were available for download at <https://climateknowledgeportal.worldbank.org/>, providing a comprehensive framework for understanding future climate trends and variability.

3.7.1 Selection of SSPs

Three distinct Shared Socio-economic Pathways (SSPs) were selected to represent a range of plausible future socio-economic conditions and greenhouse gas emission trajectories. These pathways included CMIP6 scenarios of SSP1-2.6 (low emission), SSP2-4.5 (medium emission), and SSP5-8.5 (high emission) (Figure 3.5). By selecting these SSPs, this study aimed to capture a spectrum of potential future climate outcomes, spanning from scenarios characterized by stringent mitigation efforts to those marked by high greenhouse gas emissions and limited climate action. Then, accessed climate projection data spanning from 2020 to 2100, with 0.25 degrees of spatial resolution. This data provided detailed information on future precipitation and temperature patterns and represented the mean climate anomaly products, allowing to evaluate the potential climate impacts on glacier dynamics.

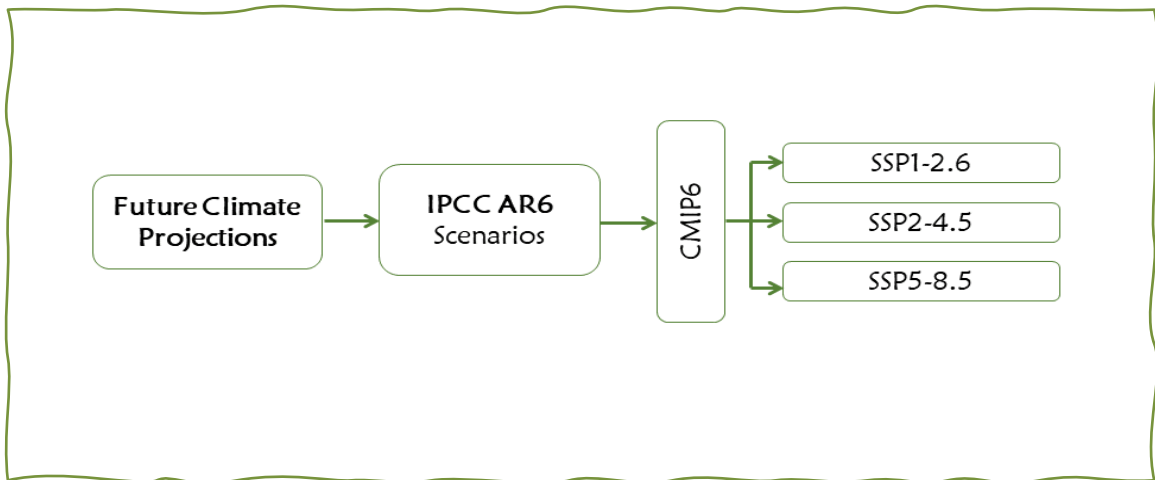


Figure 3.5: Climatic projection approach of the study.

Chapter 4

Results and Discussion

4.1 Geographic differences and glaciers dynamics

The study conducted a detailed mapping of the region of interest (ROI) to gain insights into the geographical variations and dynamics of glaciers within the selected Karakoram basins. Hypsometry, a crucial tool in glacier analysis (Cao et al., 2017), was employed to visualize the elevation distribution and area coverage of glaciers in the Hunza and Shigar basins. The analysis revealed that the majority of glaciers in these basins occupy areas ranging from less than 10 km², different size circular rings on the map (Fig. 4.1) indicating a prevalence of smaller glaciers in the region. However, a notable subset of glaciers extends over larger areas, ranging from 60 to 809 km², underscoring the diversity in glacier sizes within the study area.

The hypsometric curve, generated using SRTM DEM data from the year 2000, provided valuable insights into the topographic characteristics of the glaciated terrain. Notably, the curve exhibited a steep slope below an elevation of 5,200 meters above sea level (m a.s.l.), indicative of the rugged and high-altitude nature of the lower glacier regions. In contrast, above this altitude threshold, the slope of the hypsometric curve became gentler, suggesting a transition to more subdued topography and potentially indicating the presence of larger, flatter glacier areas. This observation highlights the complex interplay between elevation, terrain morphology, and glacier dynamics within the Hunza and Shigar basins. By integrating hypsometric data with geomorphological analysis, deeper insights were gained into the geomorphic evolution and dynamics of the glaciated regions in the study basins.

The geographical variations in glacier dynamics across the Hunza and Shigar basins reflect the complex interplay of topographic, climatic, and glaciological factors. While smaller glaciers dominate the lower elevations, larger glaciers are more prevalent at higher altitudes. This spatial variability in glacier size and distribution underscores the diverse responses of glaciers to environmental gradients, such as temperature, precipitation, and topography. Moreover, the steep

slopes observed in lower elevations suggest enhanced ice flow dynamics, whereas gentler slopes at higher elevations indicate subdued glacier behavior. Understanding these geographical differences is critical for predicting future glacier responses to climate change and assessing their implications for water resources and hazard management.

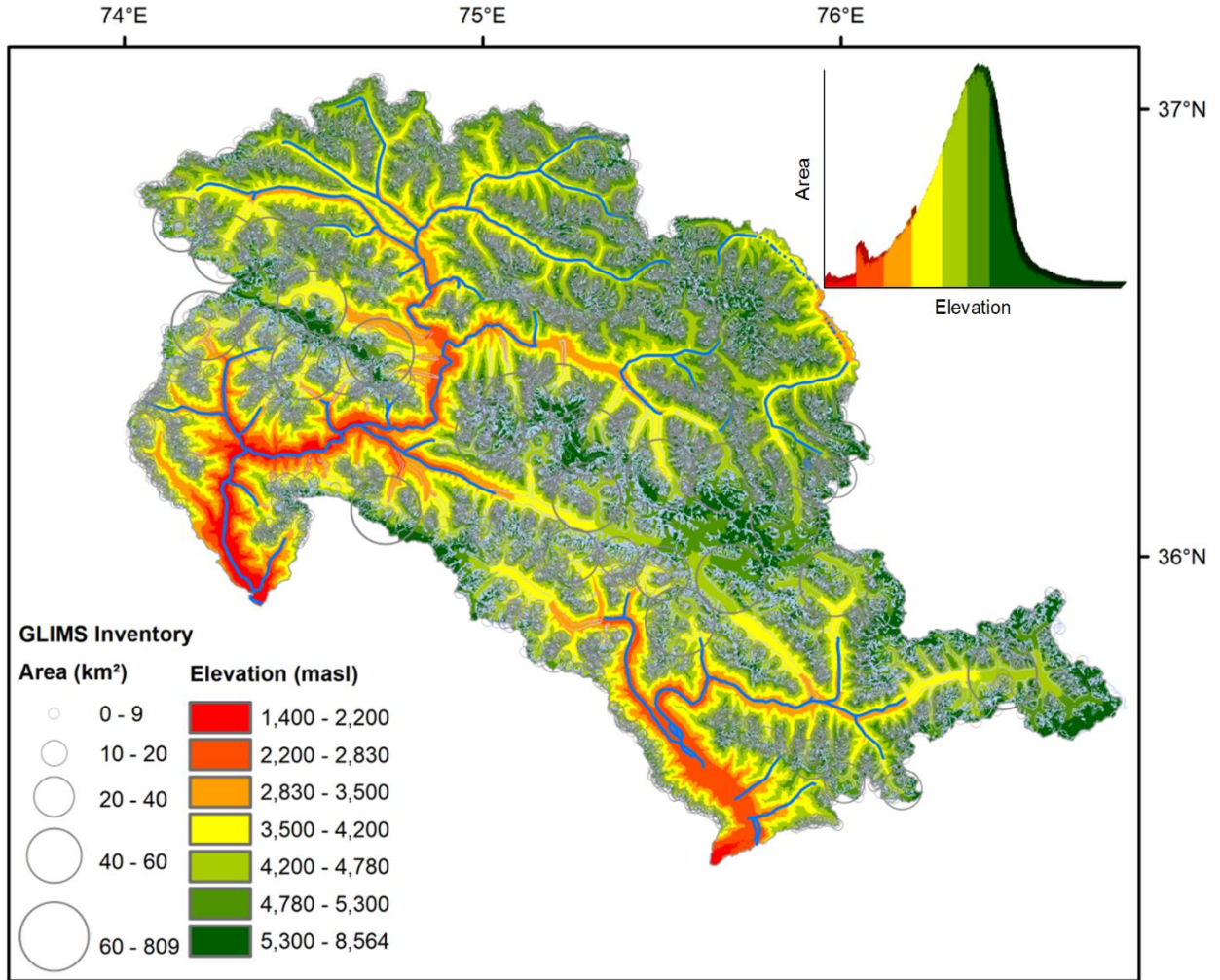


Figure 4.1: Geomorphological map of study basins.

4.2 Estimation of glaciers mass balance

Three different periods with approximately ten years gap (due to lack of DEM data continuity) were compared using cloud-free satellite images of Landsat-7ETM+ for 1991-2000, 5TM for 2001-2010, and Sentinel 2 for 2011-2022 of the selected glaciers to cover the study period from 1991-2022, subsequently set the correspond 10 m resolution to all satellite images for required

higher resolution to track the temporal changes in the glaciers outline. NDSI method used to accurately digitize the outlines of the selected glaciers, also to identify the glaciers debris covered and clean ice area through various class divisions. NDSI is a technique for mapping ice and snow cover using remote sensing images, which can effectively distinguish snow and ice from other surface features, such as clouds, shadows, vegetation, and water (Singh et al., 2021).

To evaluate the glaciers' mass balance, a geodetic technique Equilibrium line altitude (ELA) was employed. The SRTM, ALOS-PALSAR, and ASTER-14 DEMs were used for ELA calculation, which correspond to the satellite images time period. ELA-AAR and ELA-AABR are famous methods to detect the mass balance of glaciers over different time frames. DEM data was used for the elevational distribution of glaciers into different classes by extracting the contours with 50 m contour interval difference.

4.2.1 Glaciers' hypsometric analysis

The hypsometric analysis reveals varying mean elevations, with the H3 glacier in the Basin 1 exhibiting the highest mean elevation at 5,209 m a.s.l., followed by H2 at 4,631 m a.s.l., and H1 at 4,275 m a.s.l. In the Shigar Basin, the S3 glacier has the highest mean elevation at 5,307 m a.s.l., followed by S2 at 4,707 m a.s.l. and S1 at 4,668 m a.s.l. (Fig. 4.2).

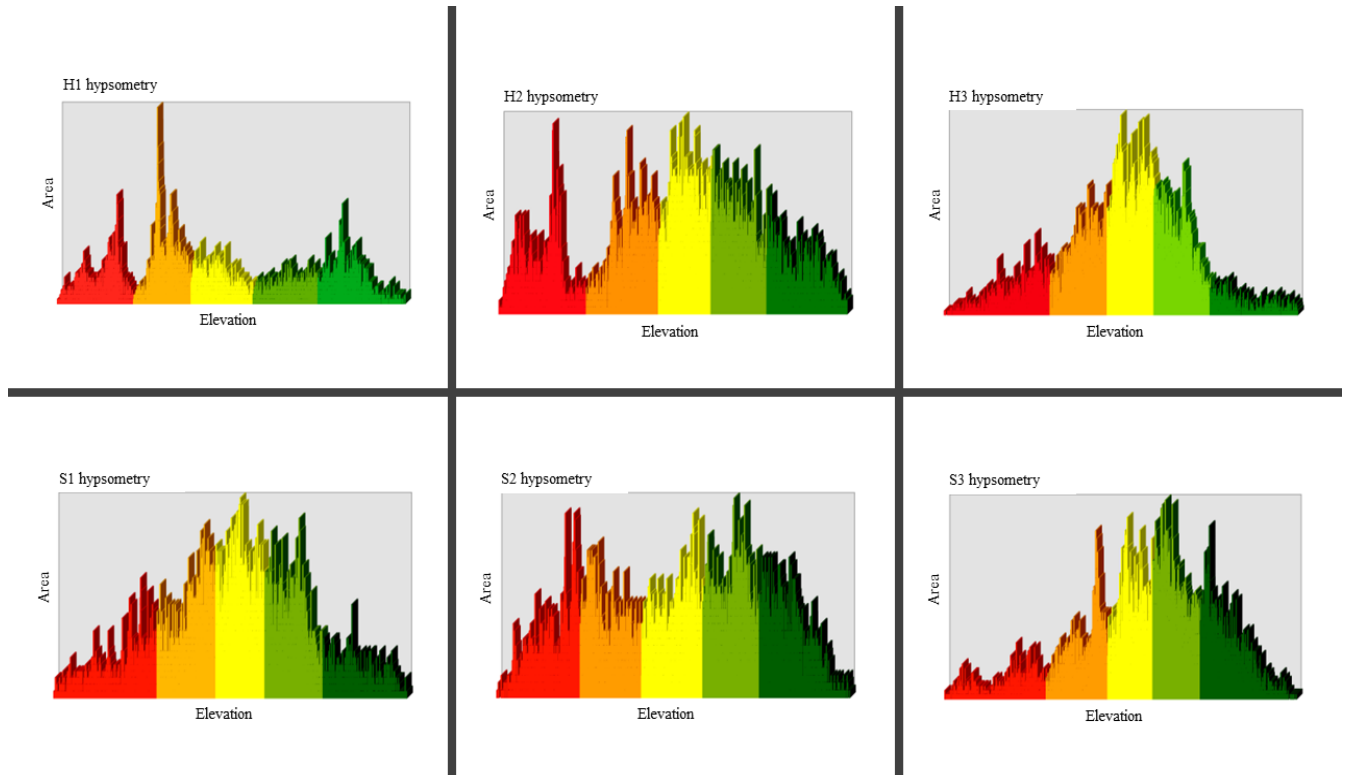


Figure 4.2: Base hypsometric curves of all studied glaciers.

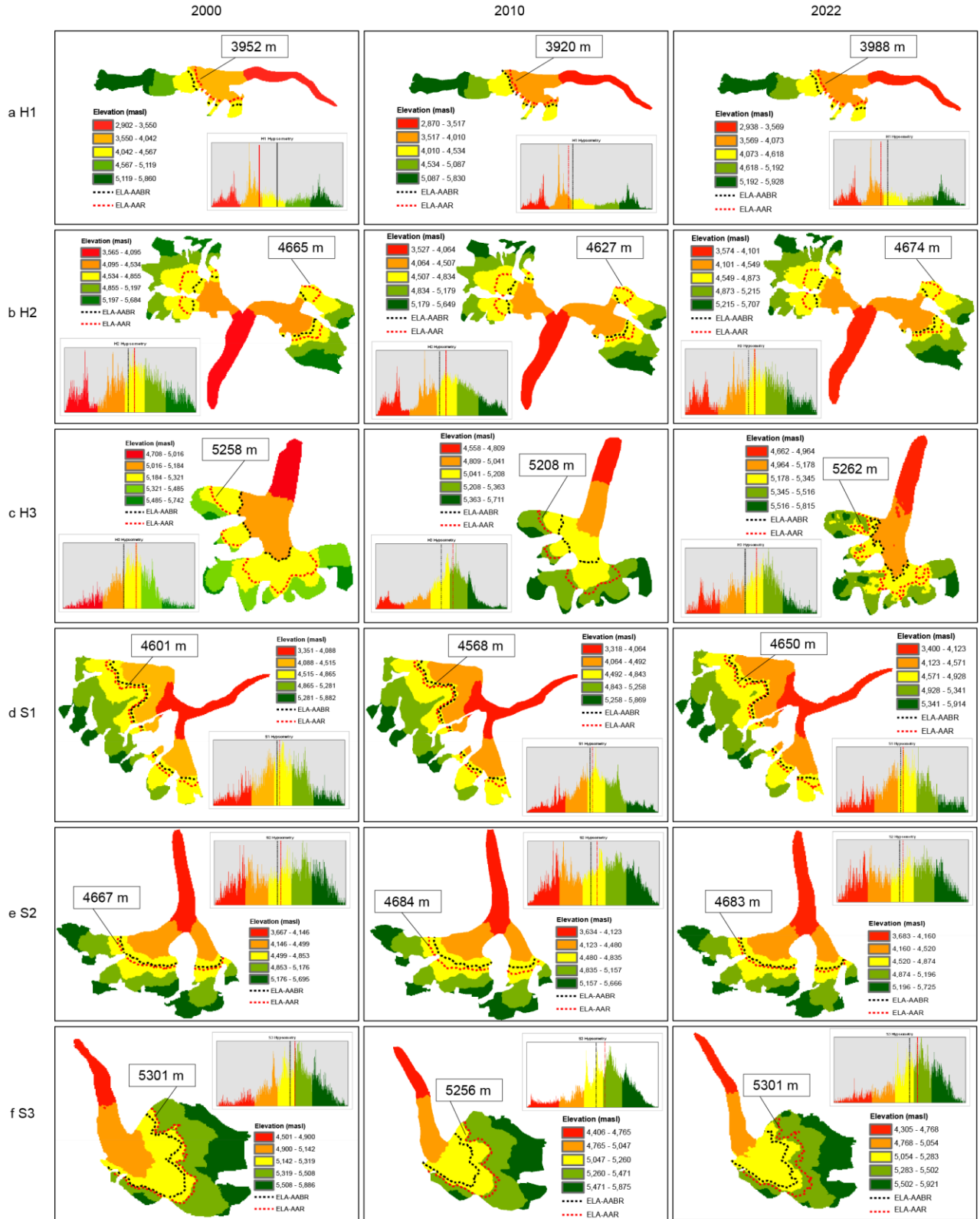


Figure 4.3: Mapped the ELA changes in selected glaciers for 1991-2022. a-f, H1 (a), H2 (b), H3 (c), S1 (d), S2 (e), and S3 (f), almost ten-year gapped ELA changes observed in columns from left to right denote the period of 1991-2000, 2001-2010 and 2011-2022 by using SRTM DEM, ALOS-PALSAR DEM and AST-14 DEM respectively. ELA-AAR denoted by red dotted line and ELA-AABR with black dotted line, similar in the hypsometry of glaciers.

4.2.2 Glacier zero mass balance

All the studied glaciers were expected to be in equilibrium with their climatic conditions based on predicted zero net balance for the ELAs, each study glacier's outlines and the reconstructed glacier area used to estimate the appropriate net-balance ELAs. The value of AAR (0.6 ± 0.05) recommended for the $>4\text{km}^2$ clean ice valley glaciers (available in Table 3.3). The correlation coefficient based linear regression model used to determine the positive or negative gain of mass balance in the Karakoram region using the AAR-ELA method (Bibi et al., 2023).

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (4.1)$$

Table 4.1: Summary of obtained ELA-AAR with 0.6 ratio & AABR with 1.56 ratio results of the selected time period in this study.

Glaciers			H1	H2	H3	S1	S2	S3
ELA (m)	1991-2000	AAR	3952	4665	5258	4601	4667	5301
		AABR	4127	4590	5183	4576	4642	5276
	2001-2010	AAR	3920	4627	5208	4568	4684	5256
		AABR	4095	4552	5133	4543	4609	5231
	2011-2022	AAR	3988	4674	5262	4650	4683	5301
		AABR	4213	4599	5187	4625	4658	5230
% Change in ELA-AAR			0.9	0.19	0.07	1	0.3	0
% Change in ELA-AABR			2.08	0.19	0.07	1.07	0.3	-0.9

The results (Table 4.2) showed a moderate positive net balance (0.53) for ELA-AAR of H1 glacier mass balance, slight positive net balance (0.18) of H2 glacier mass balance and slight positive net balance (0.07) of H3 glacier mass balance. Meanwhile, S1 mass balance has moderate positive net balance (0.59) for AAR-ELA, positive net balance (0.84) of S2 glacier mass balance and slight positive net balance (0.06) of S3 glacier mass balance. The cumulative glaciers mass balance (0.368) indicated the advancing of glaciers in the region.

Table 4.2: Summary of cumulative mass balance of glaciers used in this study with trend, through the correlation coefficient based linear regression model.

Glaciers	H1	H2	H3	S1	S2	S3
r	0.53	0.18	0.07	0.59	0.84	0.06
Cumulative Mass balance	0.368, +					

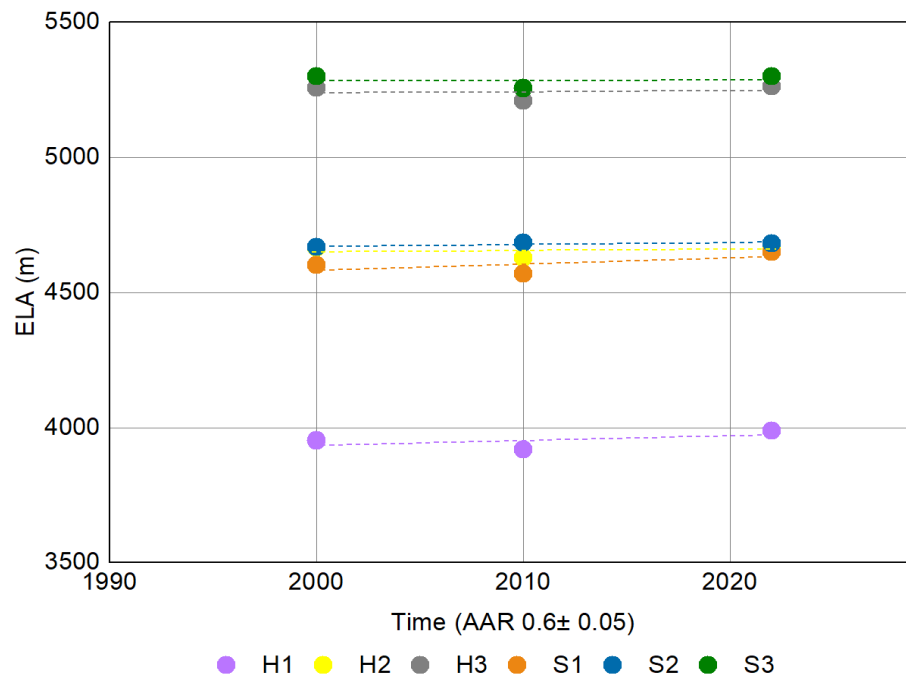


Figure 4.4: Comparison of ELA trend with time. ELA- AAR distribution across recommended ratio in the given time 1991-2022. Each glacier represented by different color filled circle as shown and identical color dotted line denotes the monotonic trend.

Based on the analysis of ELA-AAR (Fig. 4.3 and Table 4.1): the H1 glacier exhibited a slightly positive mass balance in 2000 and 2010, followed by return to a slightly negative balance in 2022. Despite fluctuations, this suggested relative stability. Similarly, the H2 glacier maintained a balanced mass state with minor variations in ELA (0.19% change). The H3 glacier also showed fluctuations in mass balance, consistent with H1 and H2, with a 0.07% change in ELA. Notably, it exceeded its snout due to a 9% increase in area (Fig. 4.5 and Table 4.3). Moreover, S1 and S2 glaciers experienced minimal ELA changes (1% and 0.3%, respectively) over the study period and S3 glacier remained steady in mass balance, while surging from the terminus with a 2% increase area. Overall, the glaciers transitioned between stable, retreat and advance states over different time spans (e.g., 1990-2010 advance to 2010- 2022 retreat) and the comparable area changes and mass balance conditions of glaciers in both basins underscored their similar positive response.

4.3 Comparison of ELA observations with AAR and AABR ratio

To improve the understanding of the mass balance condition of glaciers, ELA-AABR values were also obtained along with ELA-AAR values (Fig. 4.4 and Table 4.1), useful to compare the response of different glaciers under climate change scenarios. AABR assumes the constant ratio between the accumulation and ablation gradient, hypsometry of glaciers necessary factor to calculate it.

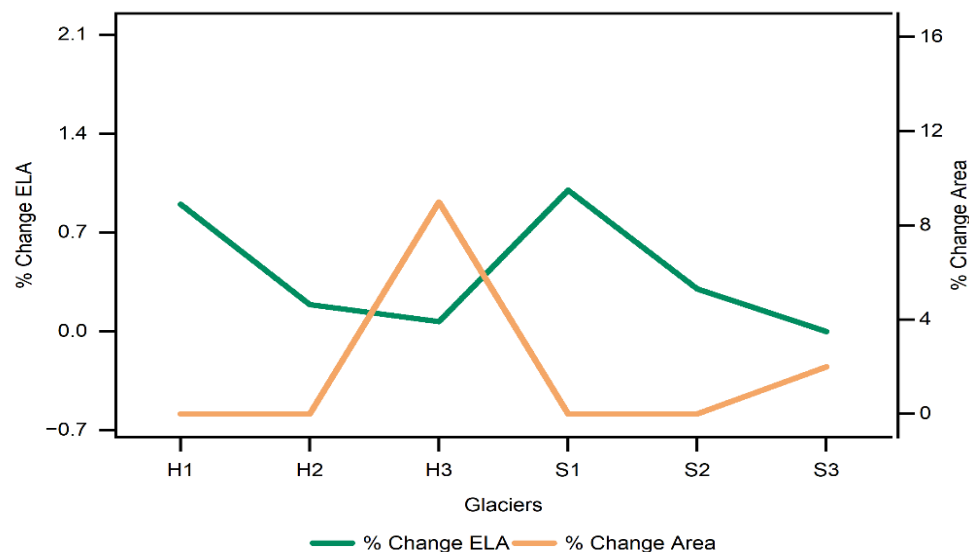


Figure 4.5: Percentage change in glaciers. Compare the calculated percentage change in the ELA and the area of glaciers during the study period.

The 1.56 AABR ratio, the recommended one for study glaciers according to scientific literature (Oien et al., 2022), was used with the 50 m contours interval and calculated by using Pellitero given GIS based with python script toolbox (same as used for AAR calculation).

Earlier discussed ELA-AAR percentage change (available in Table 4.1 and Fig. 4.6) indicated that the ELA-AABR values or percentage change showed that not much transformation experienced by glaciers during the period of 1991 to 2022. The H1, H2, H3, S1, S2, and S3 glaciers underwent 2.08, 0.19, 0.07, 1.07, 0.3, and -0.87 percentage changes respectively. This showed that all glaciers had a little decline mass or stable condition except S3 which showed slight increase mass or advancing during the study period.

Table 4.3: Summarizing the observed change in total area of glaciers in percentage and trends are given as positive, negative and null. Summary of clean ice area and debris covered ice area examined in the given time of 1991-2022.

Glaciers	Clean Ice Area (km ²)			Debris Covered Ice Area (km ²)			Change in Total Glacier Area %
	2000	2010	2022	2000	2010	2022	
H1	8.1	7.9	8.3	1.1	1.3	0.9	0,Ø
H2	11.5	11.3	12.7	1.2	1.4	0.0	0,Ø
H3	9.6	10.5	10.5	0.0	0.0	0.0	9,+
S1	11.8	11.1	11.5	1.3	2.0	1.6	0,Ø
S2	10.8	10.6	10.6	1.1	1.3	1.3	0,Ø
S3	11.4	11.4	11.4	0.0	0.2	0.2	2,+

The comparison of ELA-AAR with 0.6 ratio and ELA-AABR with 1.56 ratio observations showed that both methods can provide reasonable estimates of glacier ELAs, but with uncertainty and different degrees of accuracy. The AABR method, which considers the area and altitude

distribution (hypsoetry), had a smaller median difference (62 m, 32 m, 22 m, 43 m, 49 m, and 77 m for H1, H2, H3, S1, S2, and S3) and a narrow range of variation (± 86 m, ± 9 m, ± 4 m, ± 49 m, ± 16 m and ± 46 m for H1, H2, H3, S1, S2, and S3) than the AAR method, which considers the area ratio of the glacier, exhibiting similar variation range to AABR for all glaciers except S3 which showed no variation. The AABR method ratio also had a slight correlation with the observed ELAs, corresponding to the AAR method. The ELA-AABR method measurements explained similar glaciers response to earlier explained ELA-AAR measurements, these consistent results further supported our findings on different elevational glaciers and regions.

4.4 Calculating the clean ice and debris-covered ice area of glaciers

In addition to estimating the elevational mass balance temporally, clean ice and debris-covered ice areas calculated using satellite images from 1991 to 2022. The NDSI (Normalized Difference Snow Index) method used on images and then classified into clean ice and debris-covered ice areas (Table 4.3 and Fig. 4.6).

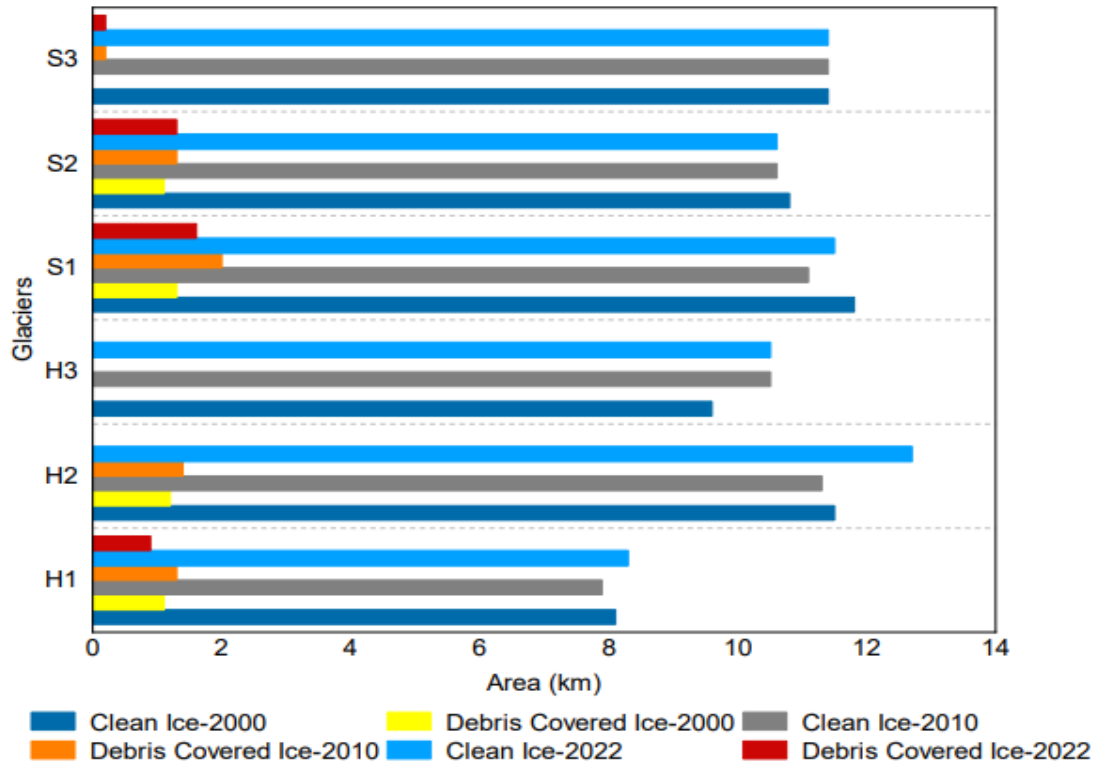


Figure 4.6: Clean ice area plot across debris covered ice area. Compare the observed ratio area of clean and debris ice in each glacier during 1991-2022, represented by the specific color given.

The percentage of total area changes of glaciers calculated, which showed that the glaciers on the higher elevations in both basins (H3 and S3) exceeded their terminus or surged. The results indicated thin debris-covered ice addition on the studied clean ice glaciers except H3 glacier in the recent decades of the twenty-first century, a narrow fluctuation in the clean ice exhibited over the study period. The presence of thin debris could behave as an insulator and reducing the ablation rate of the glaciers, to help them stay stable during warming. This change has significant impacts on the glaciers' mass balance, dynamics, and their response to climate change.

4.5 Karakoram Climate Response Anomaly (KCRA)

This study introduced the terminology “Karakoram Climate Response Anomaly” (KCRA), based on the transient long term climatic data from 1951 to 2022, it was observed that the hydroclimate of the glaciers underwent the sudden change in the beginning of the twenty first century compared to the last decades of the twentieth century, as shown in the Fig. 4.7 (a-c) with a non-uniform positive total-precipitation except H2 and S3 glaciers (which exhibited the downward trend), negative snowfall, and uniform positive temperature response.

Table 4.4: Summarizing the Mann-Kendal test and Sen's slope estimator results of the annual climatic variables data (temperature, precipitation and snowfall) of each glacier from 1951-2022 time period. P-value denotes the trend is statistically significant or not.

Glaciers	Temperature (°C)			Precipitation (mm)			Snowfall (cm)		
	tau value	p-value	Sen's slope value	tau value	p-value	Sen's slope value	tau value	p-value	Sen's slope value
H1	0.34	< 0.05	0.02	0.11	> 0.05	0.02	-0.08	> 0.05	-0.002
H2	0.26	< 0.05	0.01	-0.02	> 0.05	-0.004	-0.13	> 0.05	-0.004
H3	0.16	< 0.05	0.007	0.04	> 0.05	0.006	-0.01	> 0.05	- 0.0001
S1	0.31	< 0.05	0.02	0.05	> 0.05	0.02	-0.09	> 0.05	-0.004

S2	0.32	< 0.05	0.02	0.05	> 0.05	0.02	-0.10	> 0.05	-0.003
S3	0.29	< 0.05	0.01	-0.03	> 0.05	-0.01	-0.06	> 0.05	-0.002

The first signal of climate change is temperature, which showed weak positive response but heterogeneously and statistically significant, emerging across all glaciers, as detected by the two non-parametric tests—the Mann-Kendal test and Sen’s slope estimator (numerics available in Table 4.4).

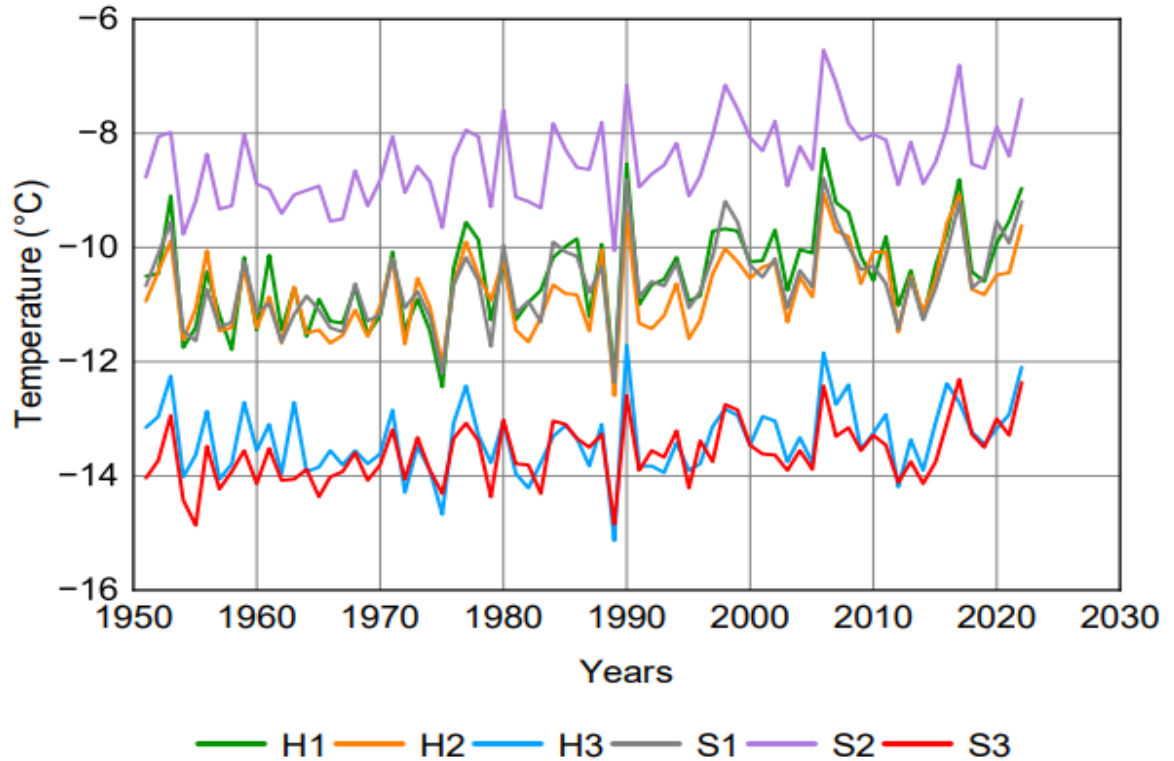


Figure 4.7: (a) Calculating the annual transient temperature response from ERA5 land monthly. Each glacier denotes by a specified colored line across ten years grid time from 1951 to 2022.

Temperature was the only statistically significant signal with varying geographical emergence across all similar elevations in the Karakoram according to literature provided in Table 3.1, due to

similar inter-annual variations at all elevations. The rise in temperature (discussed by Ali et al., 2021b, 2023) also showed an unanticipated abatement of the hydrological cycle, with total precipitation showing statistically negligible weak negative trends in the beginning of the twenty first century. All glaciers exhibited similar trends in total precipitation except H3 glacier which showed somehow uniform inter-annual variability. Snowfall generally exhibited statistically insignificant weak negative patterns, but when specifically analyzed the period from 1991 to 2022 (Appendix B), a significant negative response became evident but appeared earlier than total precipitation trend, indicating that a larger proportion of precipitation was falling as snow instead of rain, despite the temperature increase.

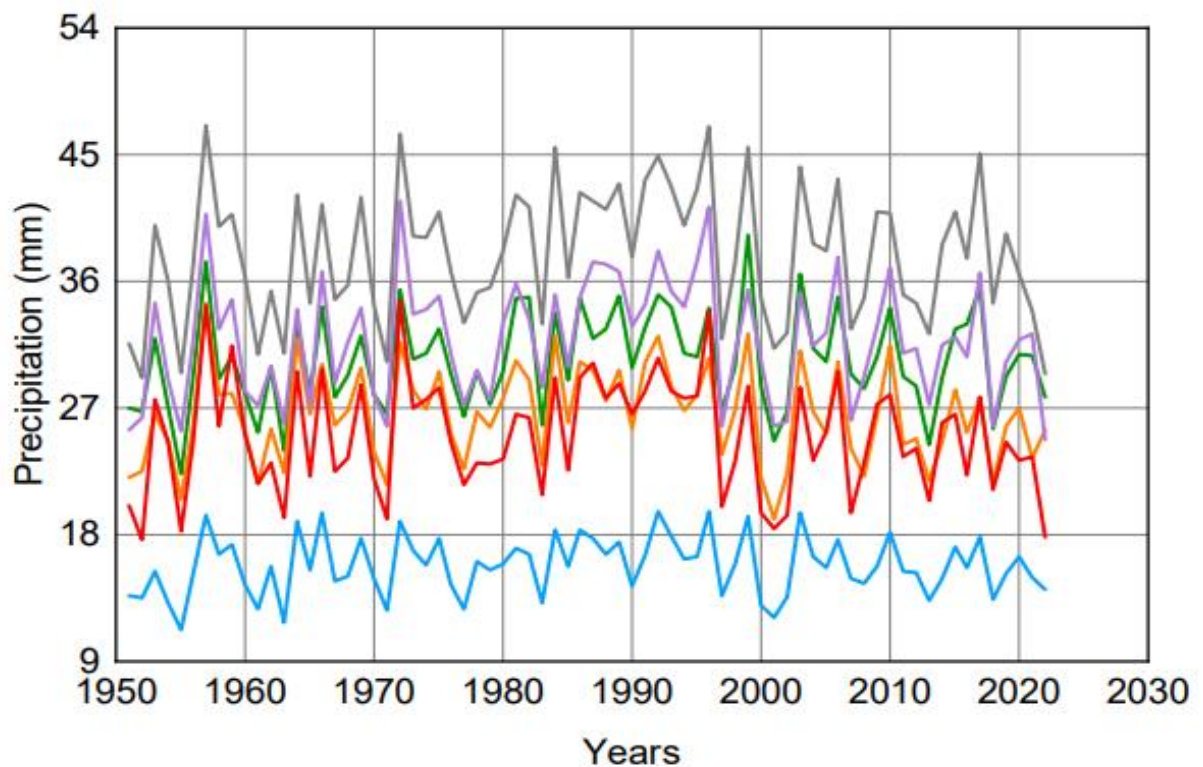


Figure 4.7: (b) Calculating the annual transient total precipitation response from ERA5 land monthly.

The cryosphere of Karakoram lacked a temporally and spatially consistent climate response. Over the other climatic variables, annual snowfall exhibited neutral to weak negative response, the statistically insignificant inter-annual variability emerging after decades. Fig. 4.7c replicated

the Karakoram glacial anomaly to the corresponding Karakoram climatic response anomaly: the annual glaciers snowfall trend was statistically weak to neutral (inconsistent) while temperature showed a contradiction. The mechanisms behind the confounding behavior of weak change in snowfall regardless of increase in precipitation and increase in annual below-freezing temperature, were difficult to interpret from annual trends alone but could be understood in the context of seasonal patterns.

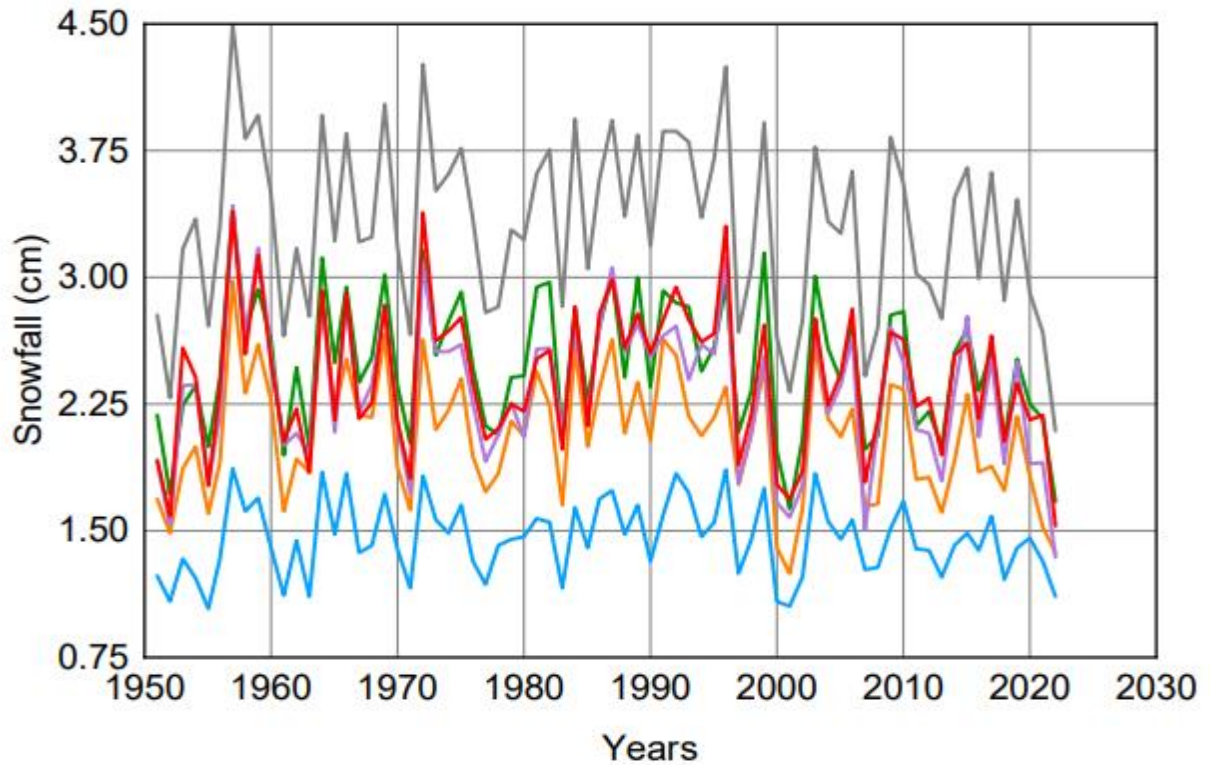


Figure 4.7: (c) Calculating the annual transient snowfall response from ERA5 land monthly.

4.5.1 Seasonal climate variability in Karakoram region

Fig. 4.8 (a-c) demonstrated how all glaciers hydroclimate annual trends were influenced by the seasonal cycle. Around the year 2000, anomalies in total precipitation and snowfall began to emerge, but statistically insignificant for snowfall and winter total precipitation mainly (available in Table 4.7 and Table 4.6 respectively). Seasonal trends usually emerged in months individually and then extended to adjacent months. All glaciers experienced weak negative snowfall anomalies in the summer and winter, but statistically insignificant (snowfall increase during October month

(Appendix C)) and mirrored the annual trends in contrast to the other variables.

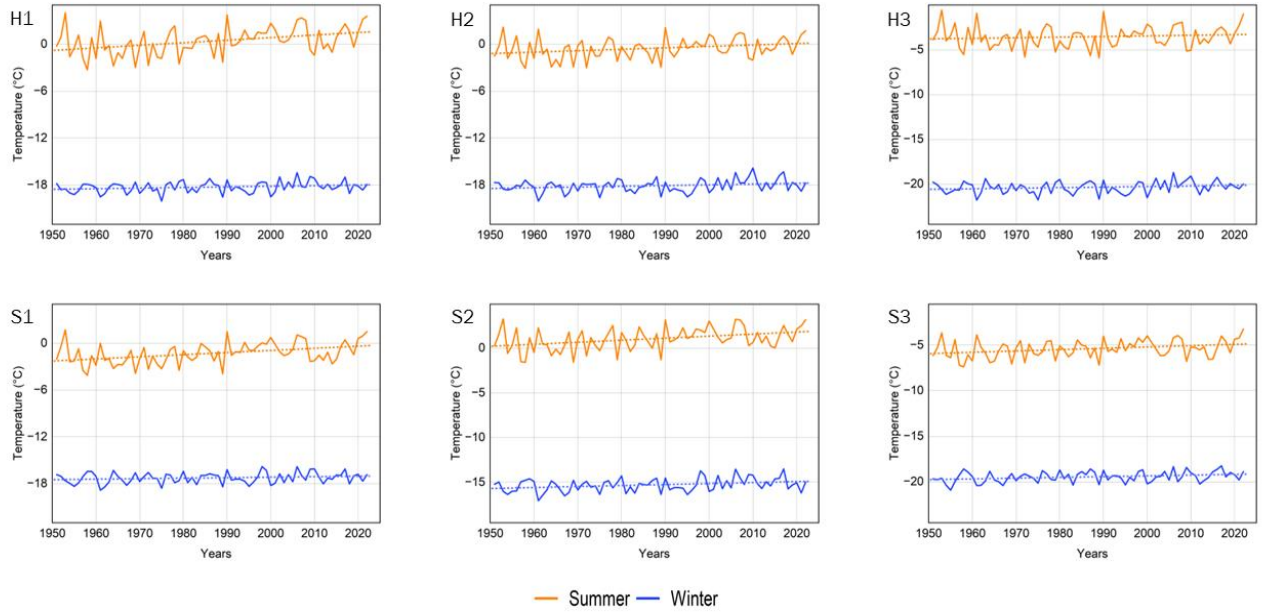


Figure 4.8: (a) Seasonal changes in temperature. ERA5 meteorological anomalies calculated to 1951-2022 of each glacier. Data was distributed in two seasons; summer (May to September) and winter (October to April) according to region's hydrological year time period, denote by bold lines and similar color dotted lines show the monotonic trend.

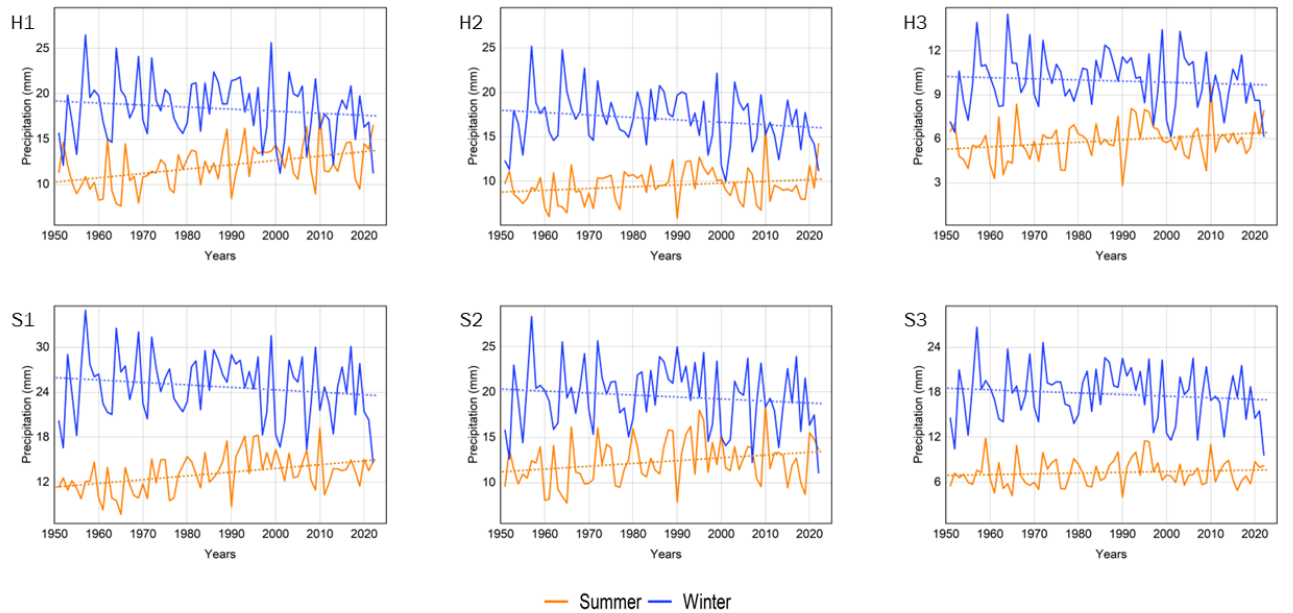


Figure 4.8: (b) Seasonal changes in total precipitation.

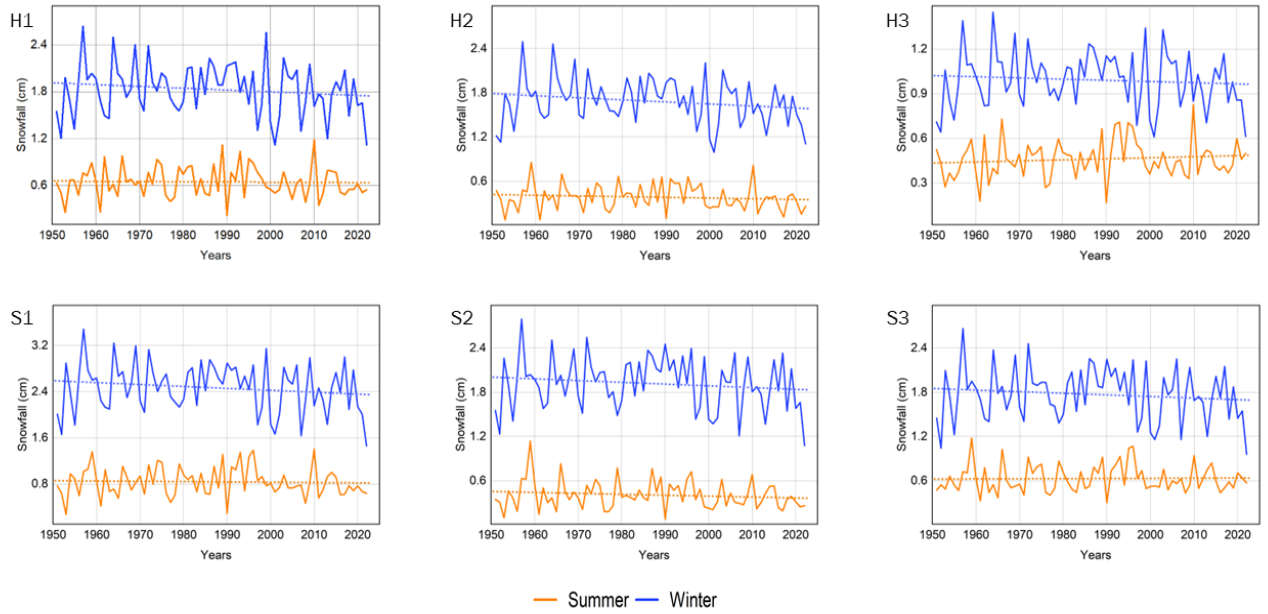


Figure 4.8: (c) Seasonal changes in snowfall.

Table 4.5: Summarizing the Mann-Kendal test with p-value and Sen's slope estimator results of the seasonal temperature data from 1951 to 2022.

	Summer Temperature (°C)			Winter Temperature (°C)		
	tau value	p-value	Sen's slope value	tau value	p-value	Sen's slope value
H1	0.29	0.0003	0.04	0.13	0.10	0.007
H2	0.19	0.01	0.02	0.14	0.09	0.007
H3	0.10	0.22	0.009	0.12	0.14	0.006
S1	0.29	0.0004	0.03	0.11	0.18	0.005
S2	0.25	0.002	0.02	0.18	0.02	0.01
S3	0.21	0.008	0.02	0.19	0.02	0.008

Similarly, total precipitation showed weak decline in winter and statistically significant rise in summer months across all elevational glaciers, which indicated gain in snowfall during summer monsoons and the adjacent months, precisely when rainfall was declining. Since snowfall dominated total precipitation annually above 3,000 m elevations and rainfall was prominent below 3,000 m elevations, the solid and liquid precipitation percentages were equal at around 3,000 m elevations. Moreover, snow contribution rised with elevation, at nearly 4,325 m elevation reaching about 75% (Singh et al., 1995). Temperature exclusively exhibited statistically significant (Table 4.5) neutral response in winter from October to April and weak enhanced in summer months across all glaciers in Fig. 4.8a.

Table 4.6: Summarizing the Mann-Kendal test with p-value and Sen's slope estimator results of the seasonal precipitation data from the period of 1951-2022.

Glaciers	Summer Precipitation (mm)			Winter Precipitation (mm)		
	tau value	p-value	Sen's slope value	tau value	p-value	Sen's slope value
H1	0.31	0.0001	0.05	-0.06	0.44	-0.01
H2	0.10	0.19	0.02	-0.11	0.16	-0.03
H3	0.16	0.05	0.01	-0.04	0.69	-0.005
S1	0.27	0.0008	0.04	-0.08	0.34	-0.03
S2	0.17	0.03	0.03	-0.06	0.45	-0.02
S3	0.09	0.25	0.01	-0.06	0.39	-0.02

Annual snowfall displayed weak negative or neutral across all glaciers under the KCRA, the overall annual cycle was statistically approximately weak neutral since its role in the seasonal cycle was comparatively smaller and balanced by increase trend in the summer snowfall (Fig.

4.8c). Moreover, temperature seasonal cycle showed neutral anomalous trend during the winter, with climatic variables not coinciding across the elevational glaciers. Seasonality exhibited by the hydroclimate was due to the periods when the glaciers temperature deviated from freezing, even though overall warming was almost constant throughout the elevations for each month. The neutral trend of temperature during the winter months did not reverse the declining snowfall trend observed across all glaciers.

Table 4.7: Summarizing the Mann-Kendal test with p-value and Sen's slope estimator results of the seasonal snowfall data from time 1951-2022.

Glaciers	Summer Snowfall (cm)			Winter Snowfall (cm)		
	tau value	p-value	Sen's slope value	tau value	p-value	Sen's slope value
H1	-0.04	0.63	-0.0006	-0.06	0.42	-0.001
H2	-0.09	0.22	-0.001	-0.12	0.14	-0.003
H3	0.06	0.48	0.0005	-0.03	0.68	-0.0004
S1	-0.05	0.55	-0.0008	-0.08	0.31	-0.003
S2	-0.09	0.27	-0.001	-0.06	0.44	-0.002
S3	0.05	0.50	0.0005	-0.07	0.38	-0.002

4.6 Determining the relationship of climate change and mass balance

As the state of glaciers is the sensitive climate change indicator, earlier estimation indicated that glaciers gained mass or remained stable in Karakoram. Further explained glaciers response to climate change, three main climatic variables (temperature, precipitation, and snowfall) used to identify the climatic conditions of glaciers.

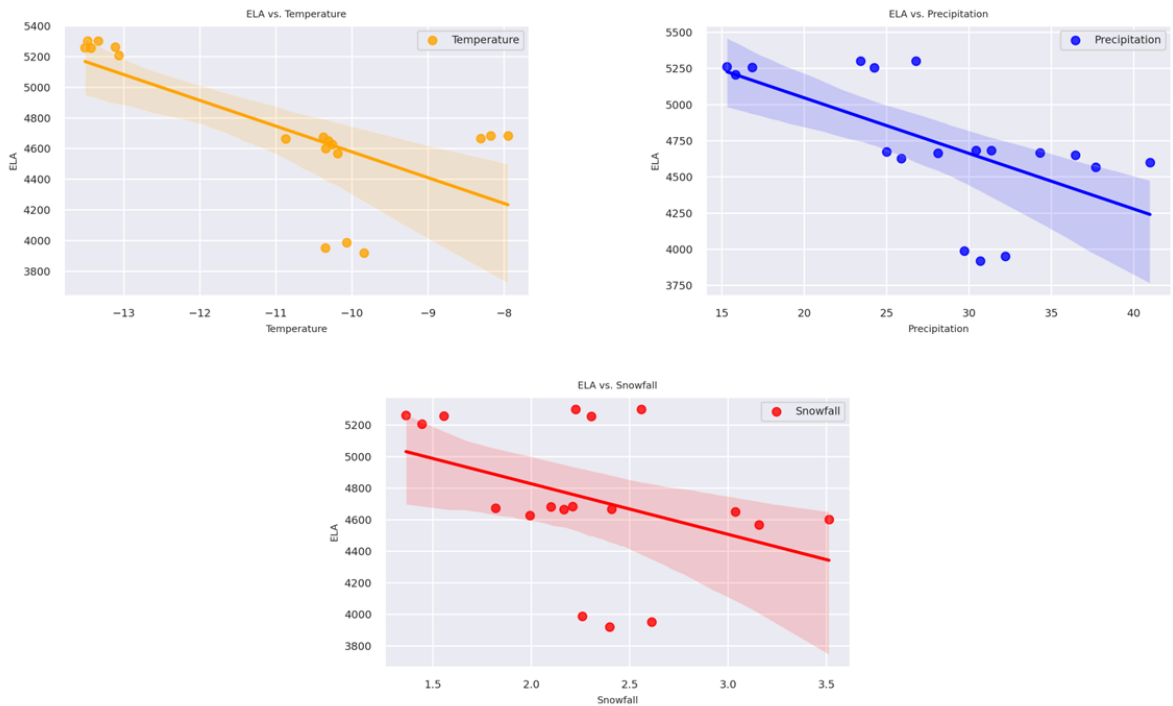


Figure 4.9: ELA response to climatic variables. Dynamic interplay between ELA and climatic variables (temperature, precipitation and snowfall) in the period of 1991-2022.

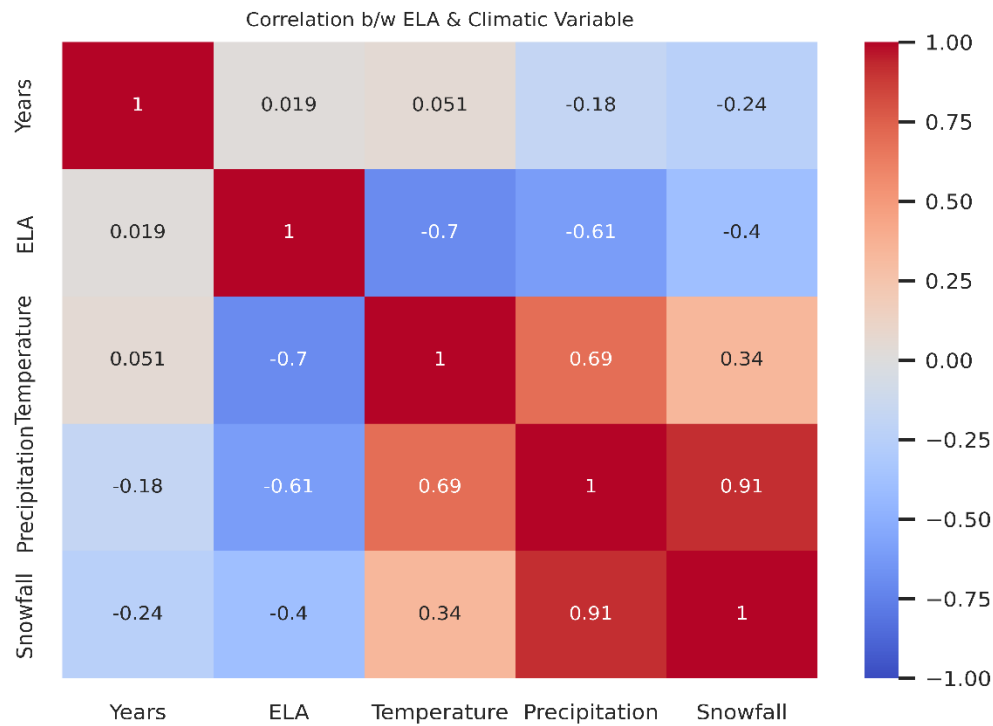


Figure 4.10: Mass balance relation with climate. Correlation between observed ELA and Climatic variables i.e., temperature, precipitation and snowfall through heatmap graph.

The two non-parametric tests; Mann-Kendal and Sen's slope estimator applied to the variables, to evaluate their temporal responses. The climatic variables showed statistically insignificant behavior (for annually and seasonally both) with time from 1951 to 2022 except the temperature which exhibited significant behavior as described earlier. The temperature demonstrated rise in annual or neutral in seasonal response, annual total precipitation showed upward trend or downward in winter season and snowfall exhibited from neutral to decreasing trend in the twenty-first century.

Notably, the smaller glaciers situated within the extreme altitudinal range of the Karakoram region, showed a statistically significant correlation with climatic patterns, with an R^2 value of 0.5, although the relation was weak (Fig. 4.9). The relationship between climate change and the mass balance of glaciers was complex and non-linear. Consequently, the Karakoram cryosphere lacked the temporally and spatially uniform response to climate change.

Table 4.8: List of observed ELA results and average of climatic variables used in this study for comparison in the given period. List of observed ELA results and average of climatic variables used in this study for comparison in the given period.

Glaciers	1990-2000				2001-2010				2011-2022			
	ELA (m)	Temperature (°C)	Precipitation (mm)	Snowfall (cm)	ELA (m)	Temperature (°C)	Precipitation (mm)	Snowfall (cm)	ELA (m)	Temperature (°C)	Precipitation (mm)	Snowfall (cm)
H1	3952	-10.4	32	2.6	3920	-9.8	31	2.4	3988	-10.1	30	2.3
H2	4665	-10.9	28	2.2	4627	-10.3	26	1.9	4674	-10.4	25	1.8
H3	5258	-13.5	17	1.6	5208	-13.1	16	1.4	5262	-13.1	15	1.4
S1	4601	-10.3	41	3.5	4568	-10.2	38	3.2	4650	-10.3	36	3.0
S2	4667	-8.3	34	2.4	4684	-7.9	31	2.2	4683	-8.2	30	2.1
S3	5301	-13.5	27	2.6	5256	-13.4	24	2.3	5301	-13.3	23	2.2

These computations demonstrated how challenging it is to identify significant climate change

across Karakoram basins. With the available gridded temporally and spatially continuous data, and the decadal high-resolution remote sensing data (in the lack of complete spatial and temporal observations records) required for all glaciers from 1991-2022, the analysis provided unique data behavior to identify climate variations in relation to mass balance of glaciers. Climatic variables were correlated on decadal basis instead of annual coincide with ELA calculations (Fig. 4.10 and Table 4.8) to highlight how the mass balance of glaciers responded to similar elevational climatic trends. Mass balance exhibited statistically significant strong negative correlation with temperature, moderate negative with total precipitation and moderate to weak negative with snowfall, indicated that the glaciers mass was not strongly affected by their climate patterns.

4.7 Aspect and slope calculations

In recent decades, thin debris covered ice measured on the snout of some glaciers, acted as insulation and protected glaciers from melting by significantly reduce the absorbed amount of solar radiations. Karakoram glaciers, mostly clean ice and a few have thin debris-covered ice on their snout during particular period of the year at certain elevations (SUPARCO, 2017).

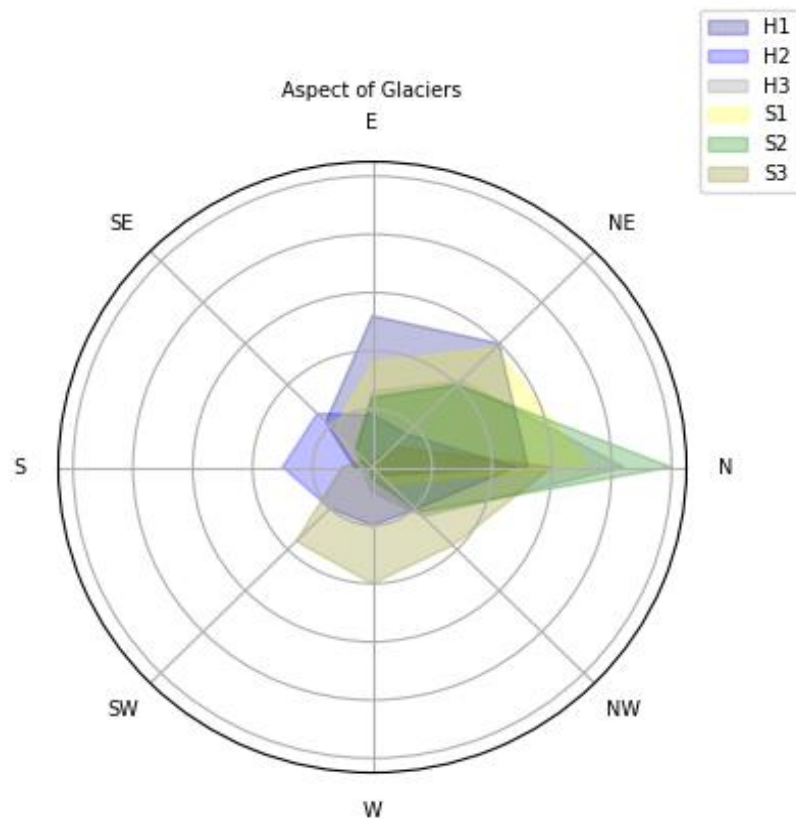


Figure 4.11: Wind-rose diagram of aspect, identify cause behind the glacier's behavior. Each glacier aspect denotes by specific color.

An analysis of glacier aspects (Fig. 4.11) and individual glaciers aspects (Appendix D), revealed that the glaciers face direction (Appendix E) mostly points towards north or northeast (this orientation implies that they accumulate less snow but retain it longer due to reduced solar radiations). Additionally, it was observed that the westerlies have shifted northward during the past few decades, the main snowfall cause over the Karakoram glaciers. The mass balance of glacier was detected by the interaction of snowfall, melting, and debris cover, which were affected by the topography, the atmospheric circulation, and the glaciers aspect.

4.8 Glaciers defying behavior towards climate patterns

Despite the similar altitudes glaciers, climate change snowfall signals exhibited different patterns across the Hunza and Shigar basins. The trend of snowfall was weakly positive correlated with temperature and strong positively correlated with precipitation in Fig. 4.10. Three weather systems influenced the area at different times of the year, two of these systems contributed to snowfall, which supported the glaciers. The monsoonal storms and air masses provided high-altitude summer snowfall, while cyclonic storms and westerly jet streams prevailed throughout the winter. It was concluded that Karakoram glaciers are influenced by a large-scale atmospheric phenomenon called the mid-latitude westerlies—a band of strong winds that carry moist air from over the oceans. The westerlies brought snow to the Karakoram mountains in the winter, most of regions' precipitation falls during this period. The snow then insulated the glaciers from melting in the summer, and increased summer precipitation (in the form of snow) might explain the observed relationship between mass balance and climatic conditions of glaciers.

These facts supported the findings and suggested that the way land shaped around the glaciers might helped them stay stable or grow, even amidst changing climate conditions (statistically insignificant snowfall pattern and increase temperature point towards the warming). Moreover, the computations of relationship between climate change and mass balance indicated that Karakoram glaciers defying behavior towards regional climate trends.

In conclusion, the Karakoram glaciers mass balance is a complex phenomenon that may combat the climatic patterns that cause glacier retreat in other regions of the world. The study showed that these mountainous glaciers are tougher than previously thought. The weak link between the climate and the glaciers mass balance challenges existing beliefs, and the north aspect of the glaciers provided more clues about why they are staying stable. The connection between the climate and the glaciers was not as strong as anticipated, this defies conventional views about glaciers and climate change relationship.

4.9 Predicted climate anomalies and glaciers behavior

To further investigate the future behavior of glaciers regarding their prevailing climate, the decadal anomalies of temperature and precipitation were analyzed, historical data from 1951-2020 based on reanalysis data and CMIP6 (Coupled Model Intercomparison Project 6) ensemble scenarios of SSP1-2.6, SSP2-4.5 and SSP5-8.5 from 2020-2100 period were examined. Fig. 4.12 demonstrated the rising temperature trend from the beginning of twenty first century in historical scenario for all glaciers and all projected SSPs scenarios indicating the rising peaks at the end of twenty first century, but the projected precipitation scenarios showed falling behavior for H3, H2 glaciers and only SSP5-8.5 for H1 glacier around 2020 to 2060 period, on the other hand, S1, S2 and S3 were projected to experience the positive precipitation trends under all scenarios, although the precipitation increase observed after 1990 in all focused glaciers.

The future scenarios of climatic data indicated that the central Karakoram (Shigar basin) and southern region of Hunza glaciers could sustain in the future under these scenarios, but the western Karakoram range (Hunza basin) glaciers might jeopardize under the future projection trends and may no longer sustain in the coming period. These findings underscore the importance of regular field observations of glaciers and the climatic variables to further or broadly understand the glacier response to climate variability.

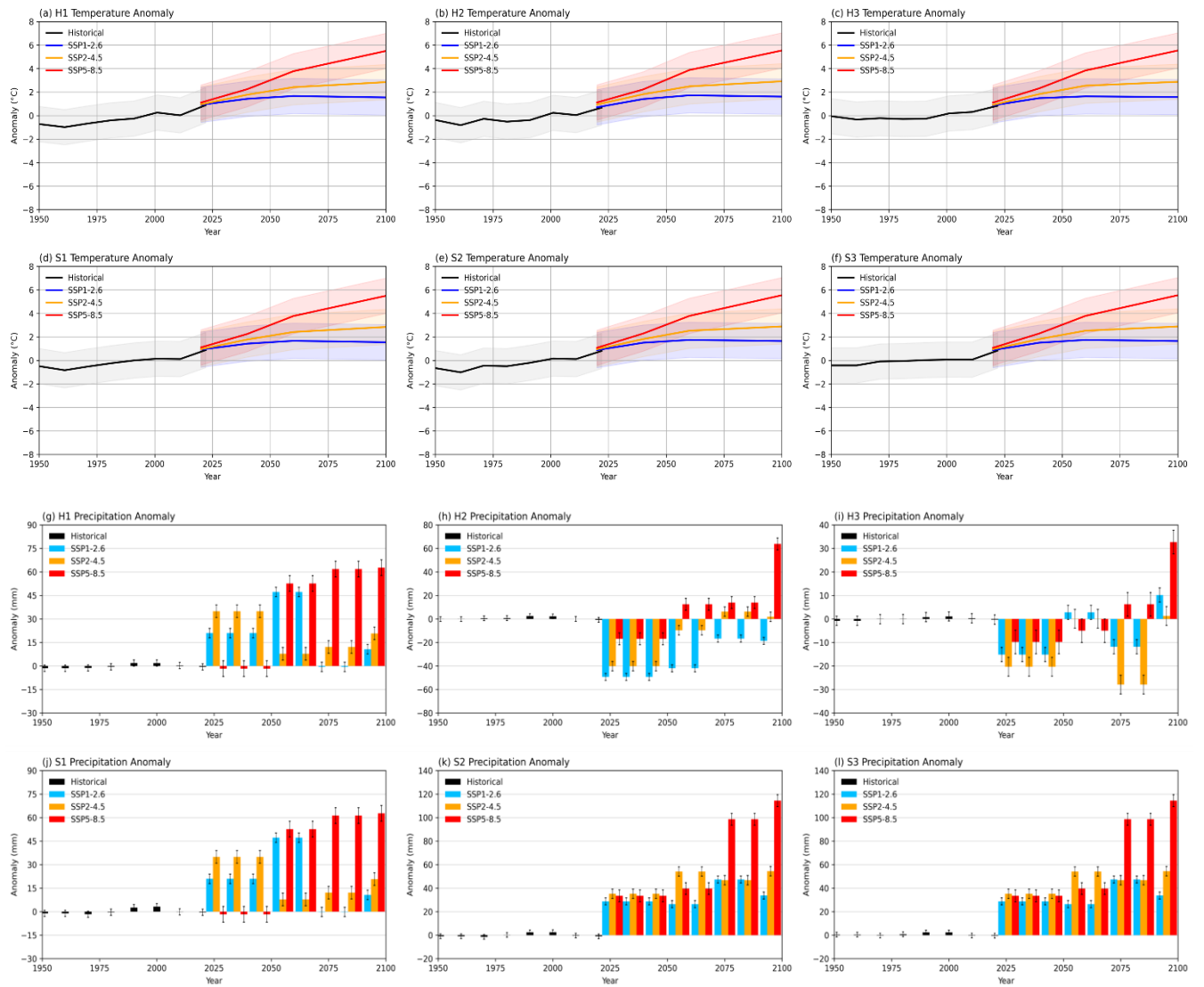


Figure 4.12: Historical and projected climate anomaly. Graphs depict decadal temperature (a-f) and precipitation anomalies (g-i), historical from 1951 to 2020 based on ERA5 reanalysis and projected CMIP6 ensemble scenarios across SSP1-2.6, SSP2-4.5, and SSP5-8.5 from period 2020 to 2100, focus on six glaciers.

Chapter 5

Conclusions

The analysis revealed consistent glaciers response in the Karakoram region through the ELA calculations and confirm the stability and positive mass balance trends over the past few decades. These findings highlight that the glaciers in this region have alternated response between periods of stability, retreat, and advance, particularly noting a general advance between 1990-2010 and a retreat from 2010-2022. The similarities in mass balance and area changes across the studied basins indicate a uniform positive response to climatic conditions. Seasonal patterns in hydroclimate play a critical role in glacier dynamics. While annual snowfall trends remain weakly negative or neutral, increased summer snowfall appears to balance out these trends. The temperature trends during winter showed a neutral anomaly, suggesting that temperature increases have not been sufficient to reverse the declining snowfall trends. This seasonality is crucial in understanding the overall glaciers' response and resilience to climate change.

The addition of thin debris-covered ice on these clean ice glaciers has been noted in the recent decades. This debris works as an insulator, reducing the ablation rate and contributing to glacier stability despite warming trends. The north aspect of glaciers might be unless noted cause behind the glaciers stability under the changing climate in the Karakoram region. The estimated anomaly is further demonstrated through the schematic diagram (Figure 5.1).

Overall, the Karakoram glaciers appear resilient despite variations in climatic patterns, but the warming trend observed over the last two decades suggests that the Karakoram glacial anomaly may not be sustain longer, with the further enhancement in warming. To assess this, historical climate anomalies were quantified using reanalysis data and projected scenarios from the CMIP6 ensemble models. Recent research indicates that Northern Hemisphere westerly winds shifting poleward due to anthropogenic forcing, a trend likely to persist with further human induced warming (Abell et al., 2021). However, there could be some regional and spatial variations exist. It seems that the Hunza basin (western part of Karakoram region) is experiencing rising projected temperatures across all SSPs coincide with drier conditions and reduced precipitation, particularly

pronounced during 2020–2060. Speculatively, the Karakoram glacial anomaly may dissipate in the western part of the range in the coming decades.

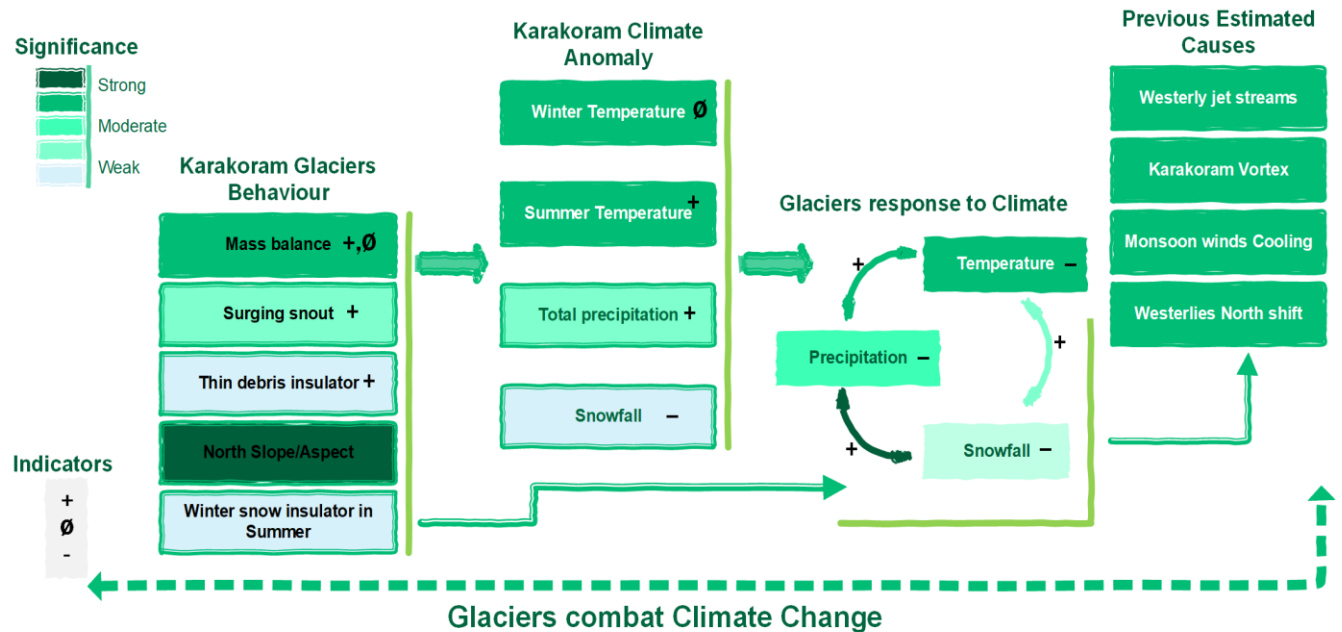


Figure 5.1: Schematic diagram of estimated anomaly. The statistical significance of every phenomenon related to anomaly is estimated from high to low. Increase, neutral and decrease trend of study variables indicate by specific signals and cite literature explanation of anomaly.

5.1 Recommendations

- I. It is imperative to establish ongoing field monitoring of climatic variables and glacier mass balance in the Karakoram region. This will provide more precise data and improve the understanding of glaciers response to climate change.
- II. Further studies should be conducted to focus on the debris cover role in glacier dynamics. Understanding how debris impacts ablation and mass balance will be crucial in predicting future changes in glacier behavior.
- III. Utilize high-resolution climate projection models to refine predictions for the Karakoram region. This includes assessing the impact of regional climatic changes and anthropogenic influences on glacier stability.

- IV. Policymakers, scientists, and the public should be informed about the unique resilience of the Karakoram glaciers. Awareness programs and policy decisions should consider the potential foreseeable changes in glacier behavior and their implications for water resources and regional ecology.
- V. Encourage the similar interdisciplinary research combining glaciology, climatology, and remote sensing. This integrated approach will provide comprehensive insights into the intricate interactions between climate and glaciers in the Karakoram region.

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Appendices

Appendix A: GEE code sample of SRTM dem data

```
// Define the region of interest as a bounding box
var roi = ee.Geometry.Rectangle([xmin, ymin, xmax, ymax]); // Define the coordinates of the
bounding box

// Load SRTM DEM dataset
var dem = ee.Image('USGS/SRTMGL1_003');

// Clip the DEM to the region of interest
var demClip = dem.clip(roi);

// Define visualization parameters
var demVis = {
  min: 0,
  max: 4000,
  palette: '000000, 0000FF'
};

// Display the clipped DEM on the map
Map.addLayer(demClip, demVis, 'SRTM DEM (clipped)');

// Export the clipped DEM to Google Drive
Export.image.toDrive({
  image: demClip,
  description: 'SRTM_DEM',
  folder: 'SRTM',
  scale: 30, // Adjust the scale as per your requirement
  region: roi,
  crs: 'EPSG:4326', // refers to the coordinate reference system (CRS) known as WGS84
  maxPixels: 8e8 // Adjust the maximum pixel count as per your requirement
});

// Acquisition date of DEM
var metadata = dem.getInfo().properties;
console.log(metadata);
```

Appendix B: Annual snowfall data results

Table: Summarizing the Mann-Kendal test with p-value and Sen's slope estimator results of the annual snowfall data specifically from 1991 to 2022.

Glaciers	Snowfall (cm)		
	tau value	p-value	Sen's slope value
H1	-0.254	< 0.05	-0.0161
H2	-0.286	< 0.05	-0.0171
H3	-0.274	< 0.05	-0.0093
S1	-0.294	< 0.05	-0.023
S2	-0.246	< 0.05	-0.018
S3	-0.266	< 0.05	-0.016

Appendix C: Four months climate anomaly of hydrological year

The climate anomalies of the hydrological year were analyzed by considering four months separately included April (ending month of the winter), May (starting month of the summer), September (last month of the summer and also hydrological year) and October (starting month of the winter and also first month of the hydrological year). These months were referred according to the Karakoram atmospheric conditions (referred here from literature). The climatic variables (temperature, precipitation and snowfall) anomalies of these four months from 1951 to 2022 of each studied glacier exhibited through figures below.

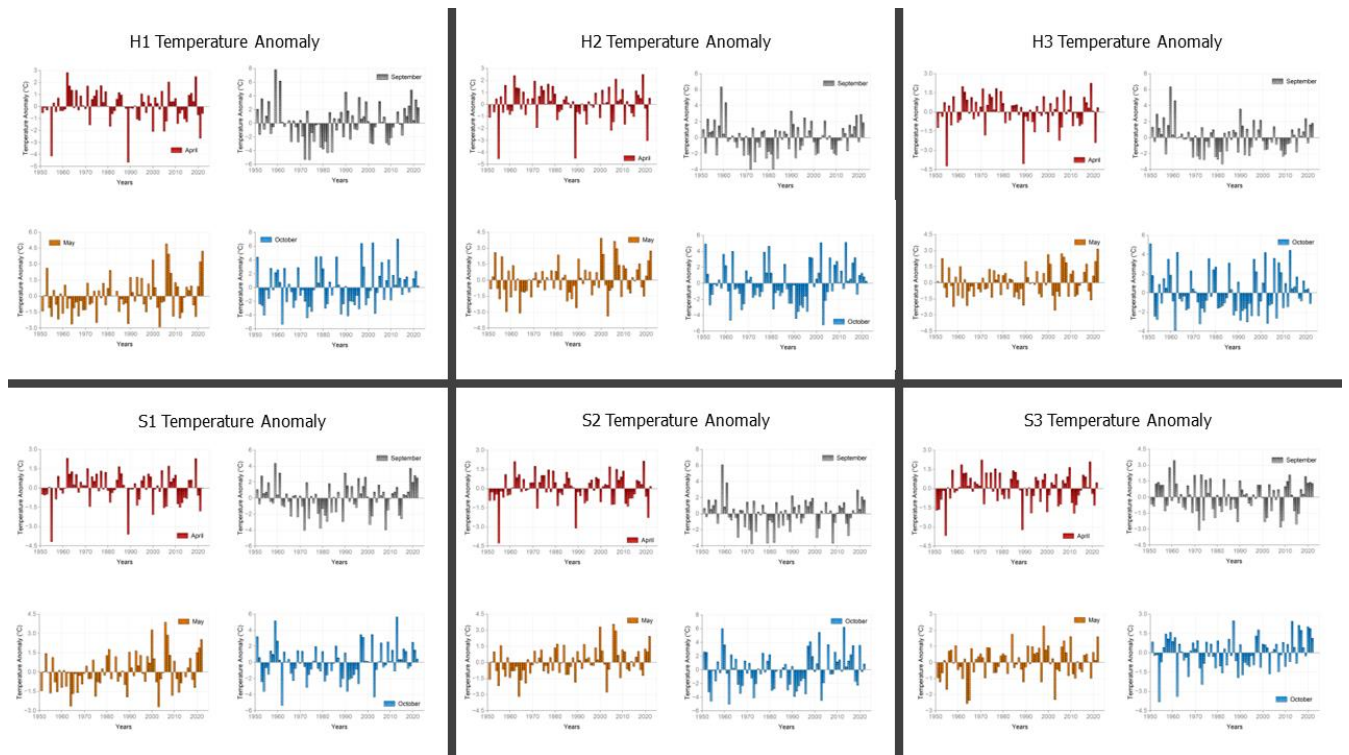


Figure: Four months temperature anomaly of each glacier from 1951 to 2022.

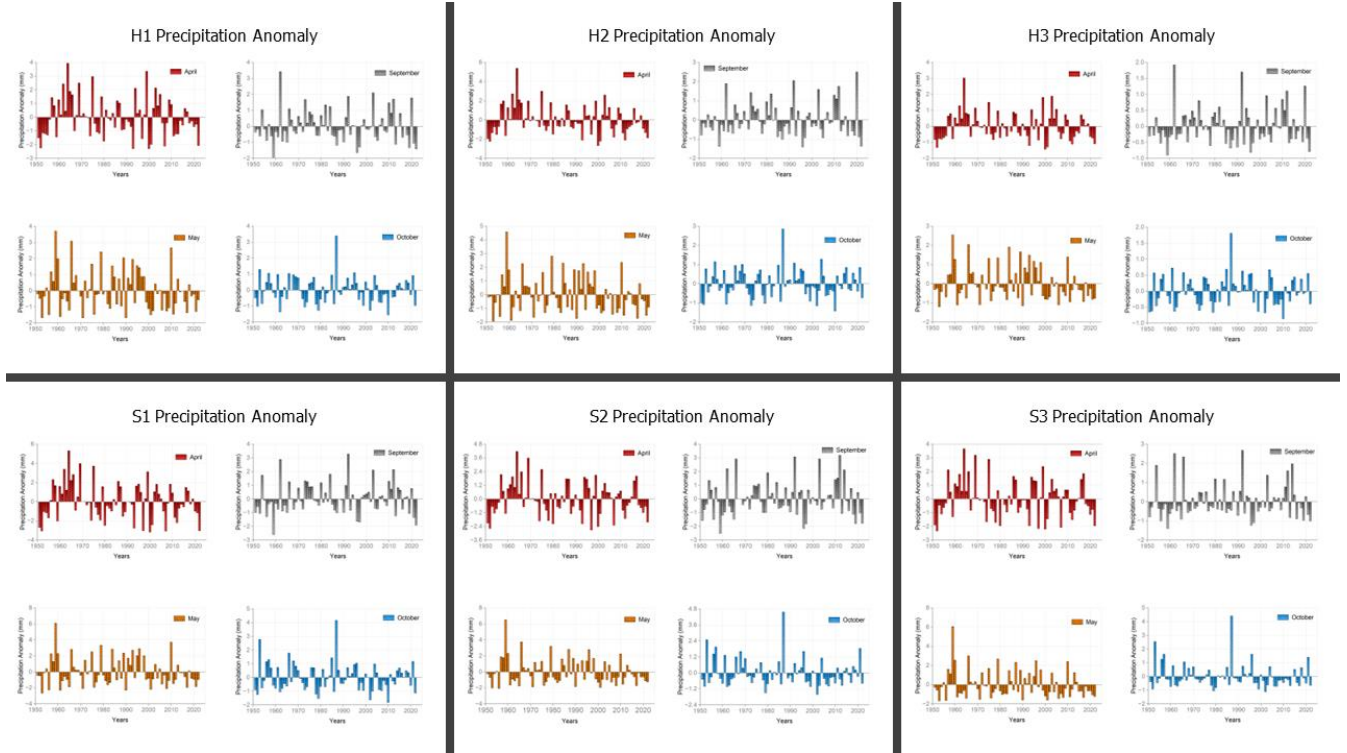


Figure: Four months precipitation anomaly of each glacier from 1951 to 2022.

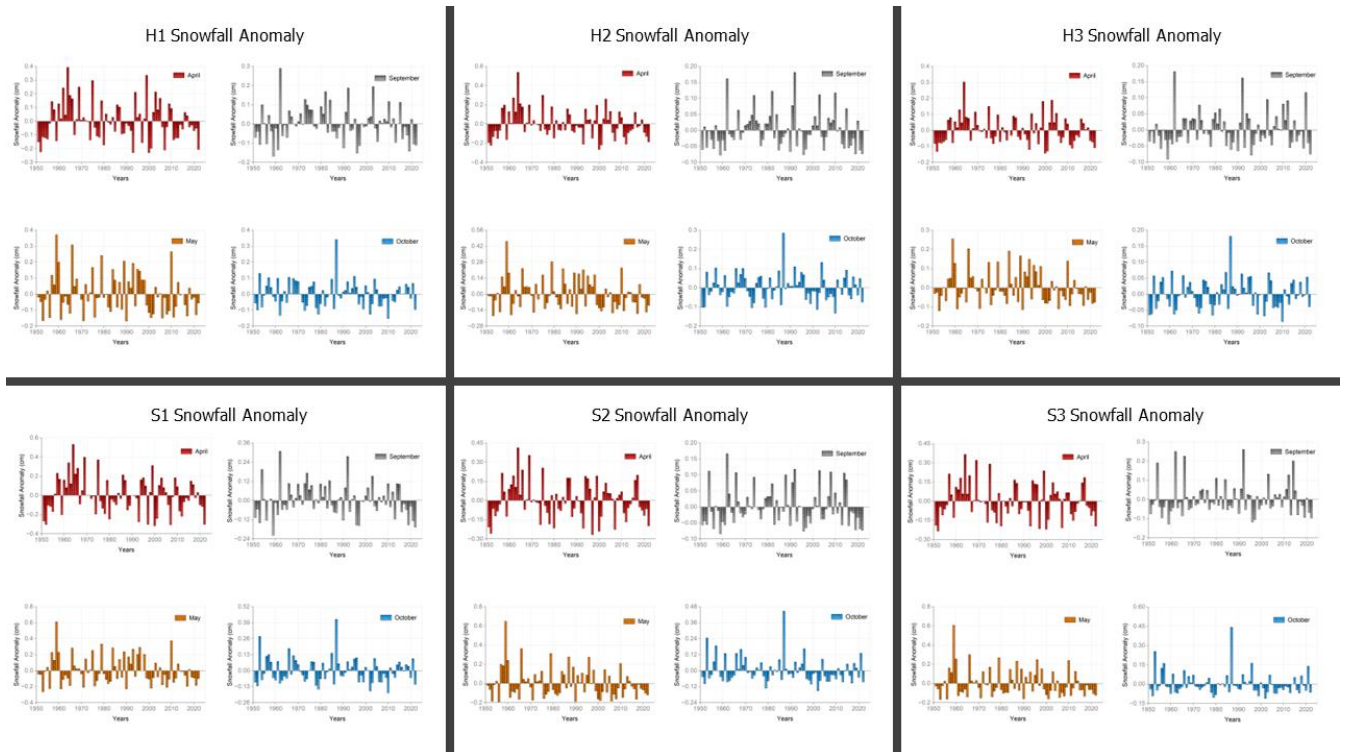


Figure: Four months snowfall anomaly of each glacier from 1951 to 2022.

Appendix D: Aspect of studied glaciers

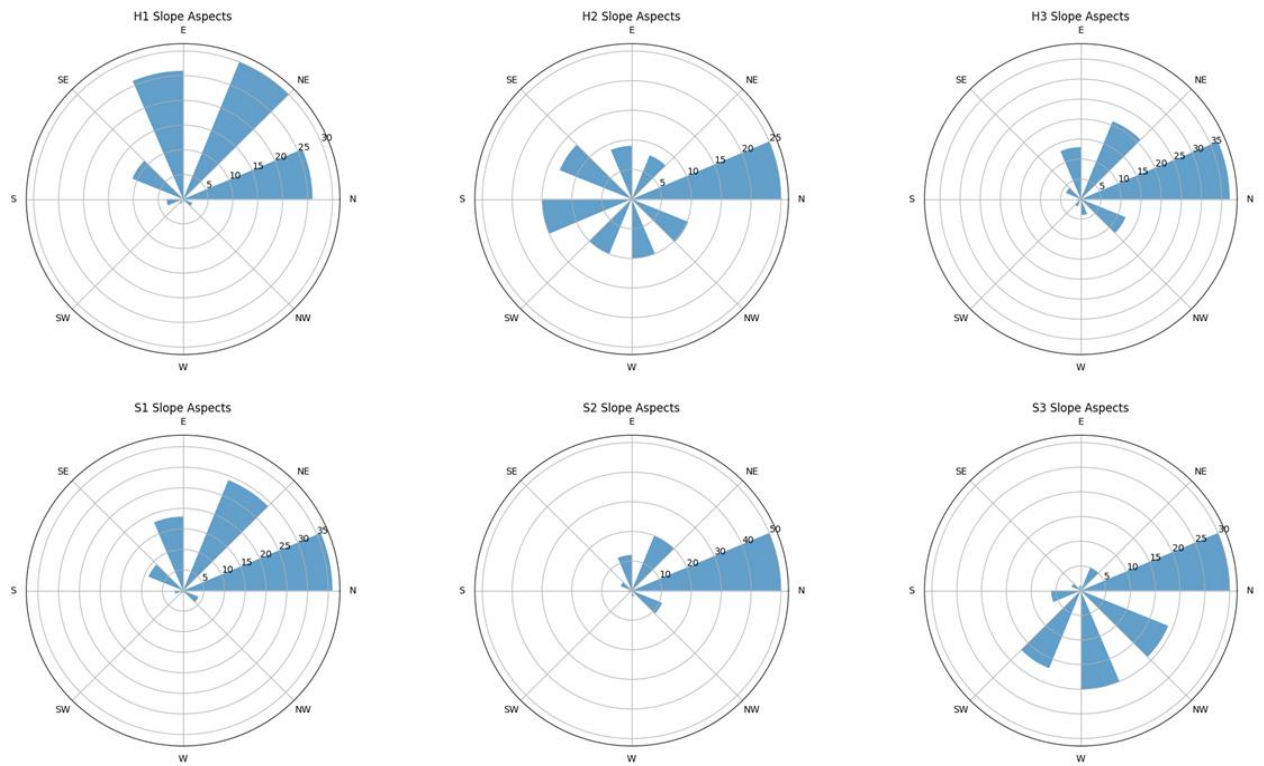


Figure: Wind rose diagram showing individual glacier aspects.

Appendix E: Studied glaciers face direction

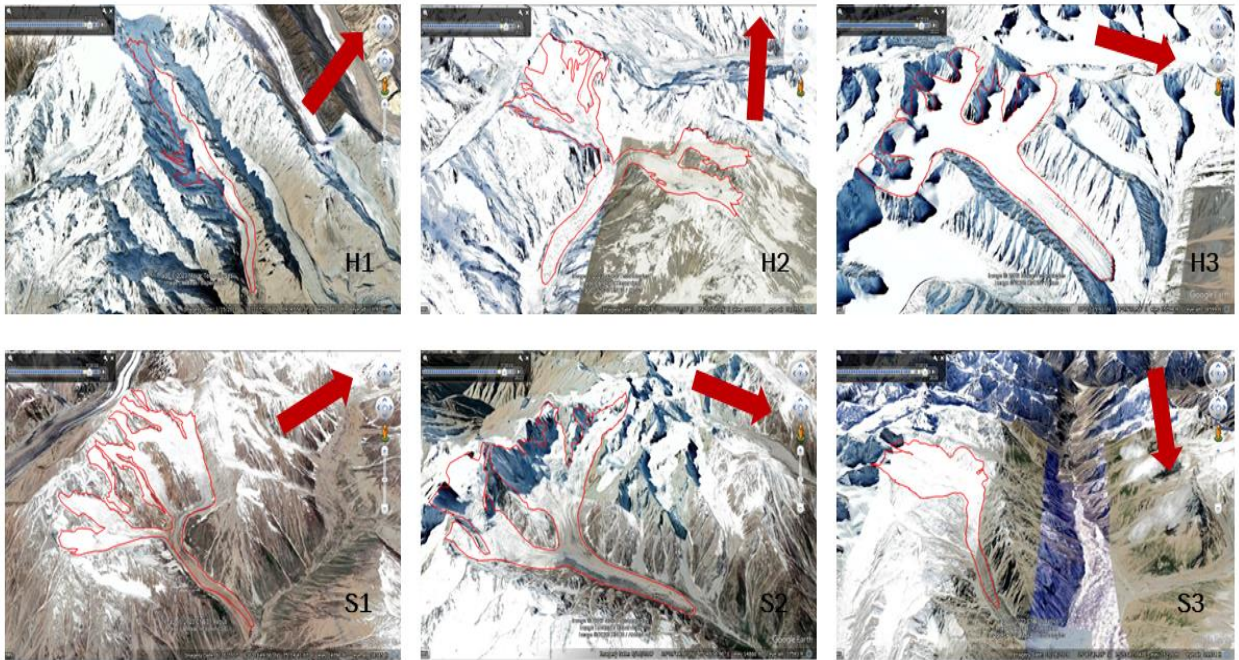


Figure: Studied glaciers terminus direction shown through google earth recent snapshots. North direction denoted by red arrow.